Lab 1 – Acquiring Digital Evidence with Open-Source Tools

1. Introduction to FTK Imager

FTK Imager is a GUI-based tool used to make forensic images of Flash Drives, Disks, Partitions, and Volatile Memory. You can preview the contents of forensic images stored on the local machine or on the network drive. It can also mount an image and leverage the power of Windows File Explorer to see the content on the original drive as is.

AccessData offers FTK Imager as open-source software that creates accurate copies of original evidence without making any changes to it. The original evidence image remains the same and enables us to copy data at a much faster rate, which we can preserve and analyze further.

The FTK imager provides an inbuilt integrity checking function which generates a hash report that helps in matching the hash of the evidence before and after creating the image of the original evidence.

2. Installing FTK Imager

FTK Imager is downloaded from the website: https://accessdata.com/product-download/ftk-imager

Simply visit the website and click on the Download button. This will take you to another website to fill in data. You don't have to provide your real information. Use services like 33mail or 10minutemail to create a dummy account. It does not ask for verification.

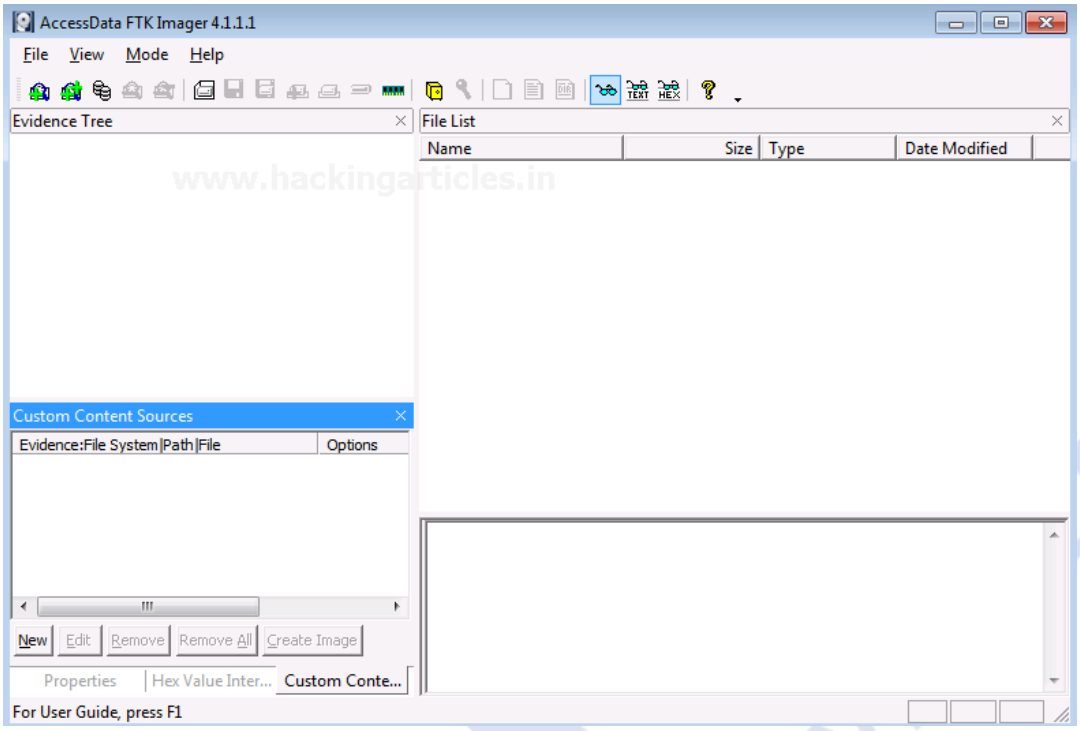
Once you launch the downloaded application, it will install FTK Imager into the path that you choose. Follow the steps displayed by the wizard and you will have FTK setup.

3. Creating a Forensic Image

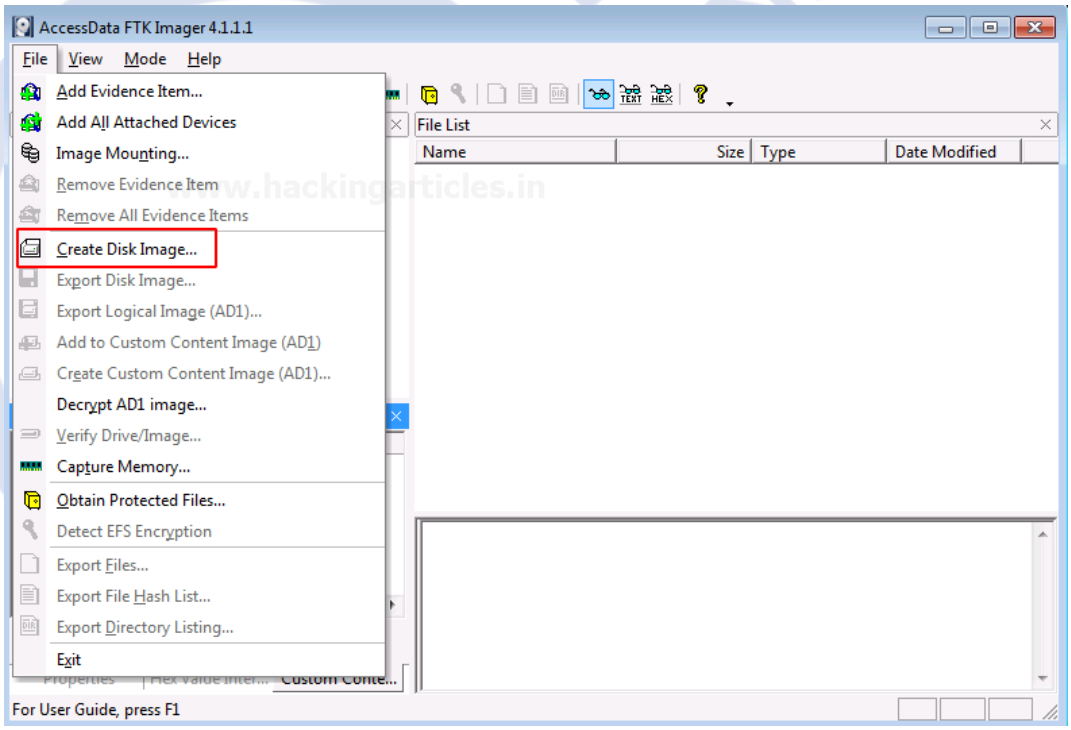
Forensic Imaging is one of the most crucial steps in digital forensic investigation. It is the process of making an archival or backup copy of the entire hard drive. It is a storage file that contains all necessary information to boot to the operating system. However, you need to apply this imaged disk to the hard drive to work. You cannot restore a hard drive by placing the disk image files on it, as you need to open and install them on the drive using an imaging program. A single hard drive can store many disk images. You can also store disk images on flash drives with larger capacity.

3.1 Launching FTK Imager and Creating Disk Image

Open FTK Imager by AccessData after installing it, and you will see the window pop-up which is the first page to which this tool opens.



Now, to create a Disk Image, click on **File > Create Disk Image**.

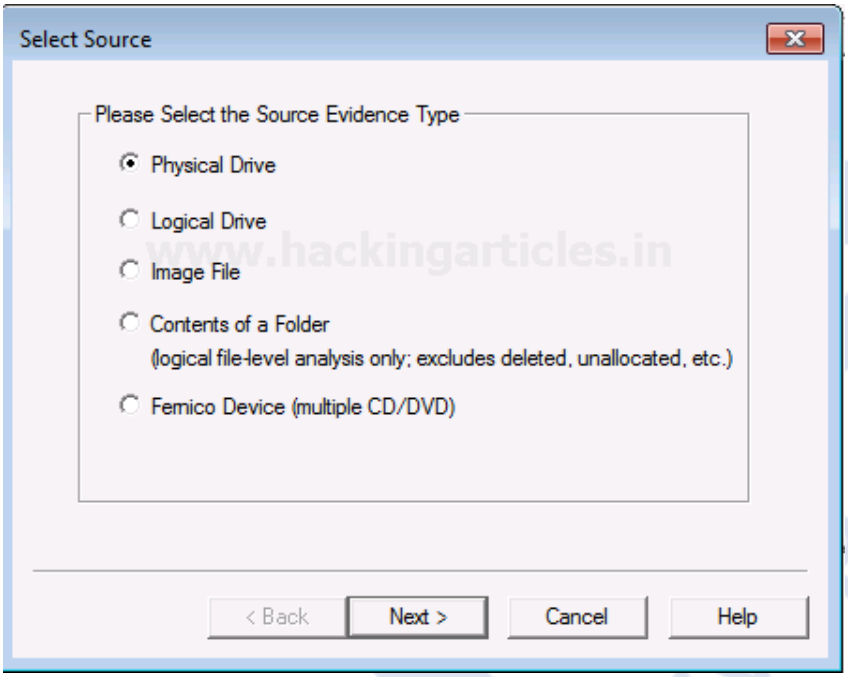


You can choose the source based on the drive you have. It can be a physical or a logical drive depending on your evidence.

3.2 Understanding Physical and Logical Drives

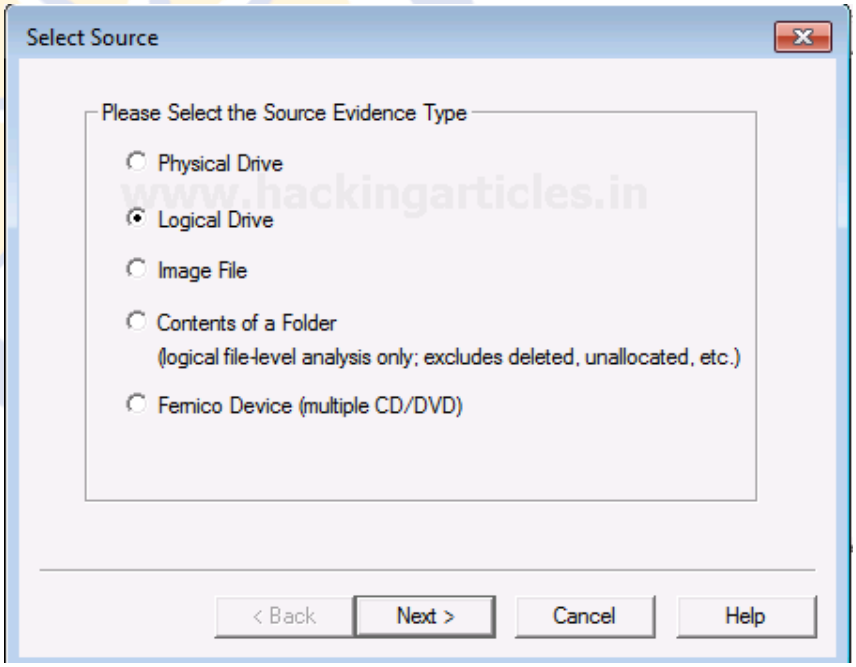
Physical Drive

A device uses a Physical Drive as the primary storage hardware or component to store, retrieve, and organize data. On Linux, the equivalent would be /dev/nvme0n1.

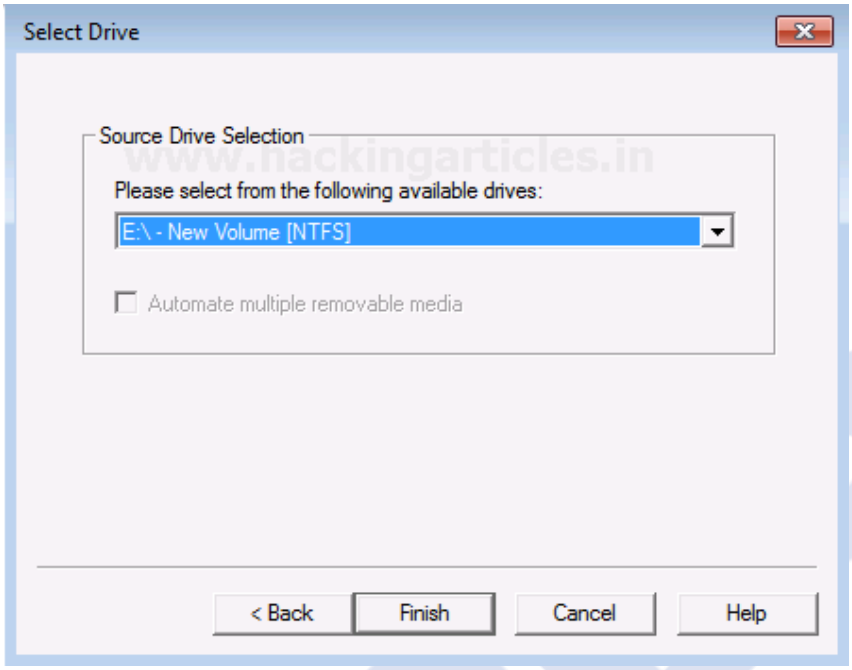


Logical Drive

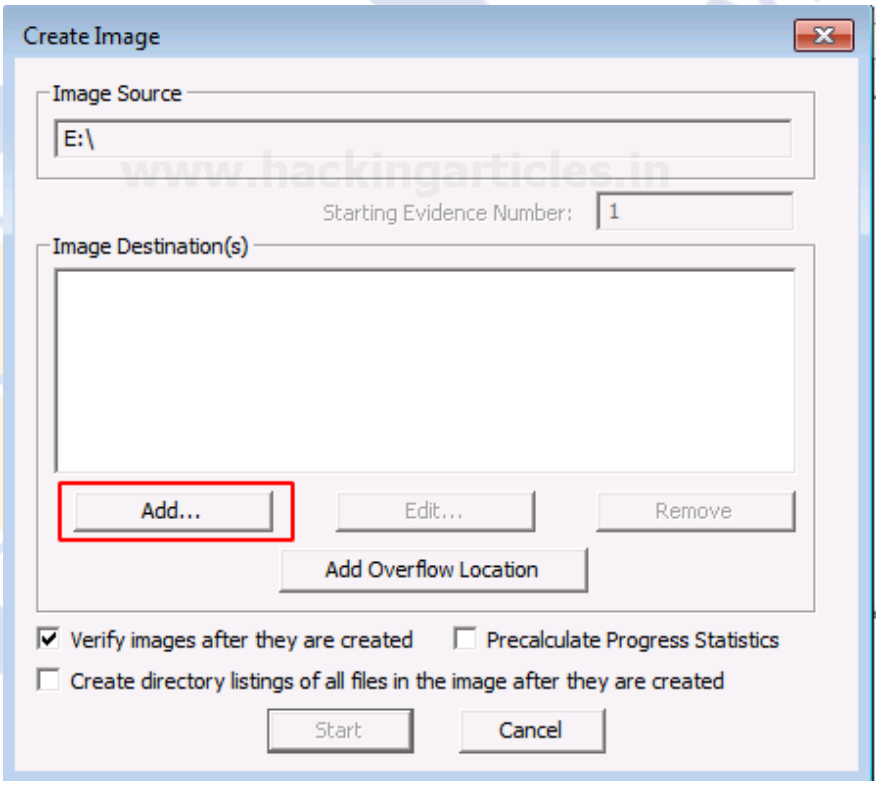
A physical hard disk generally creates a Logical Drive as a drive space. A logical drive has its parameters and functions because it operates independently. These are the partitions we create like /dev/sda1 or /dev/sdb1.



Choose the source of your drive that you want to create an image copy of.



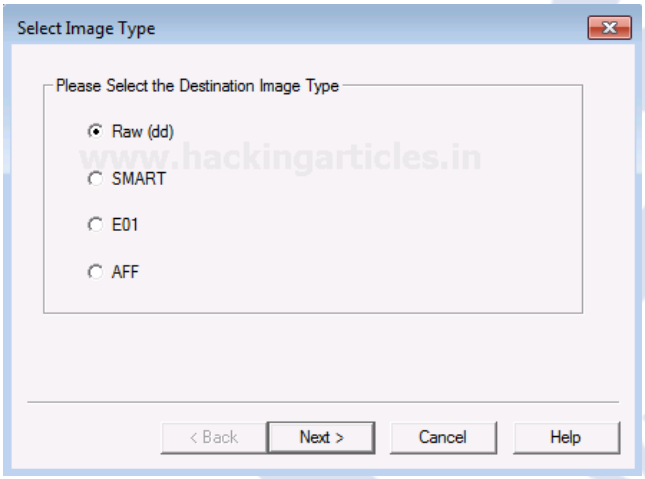
Add the destination path of the image that is going to be created. From a forensic perspective, it should be copied to a separate hard drive and multiple copies of the original evidence should be created to prevent loss of evidence.



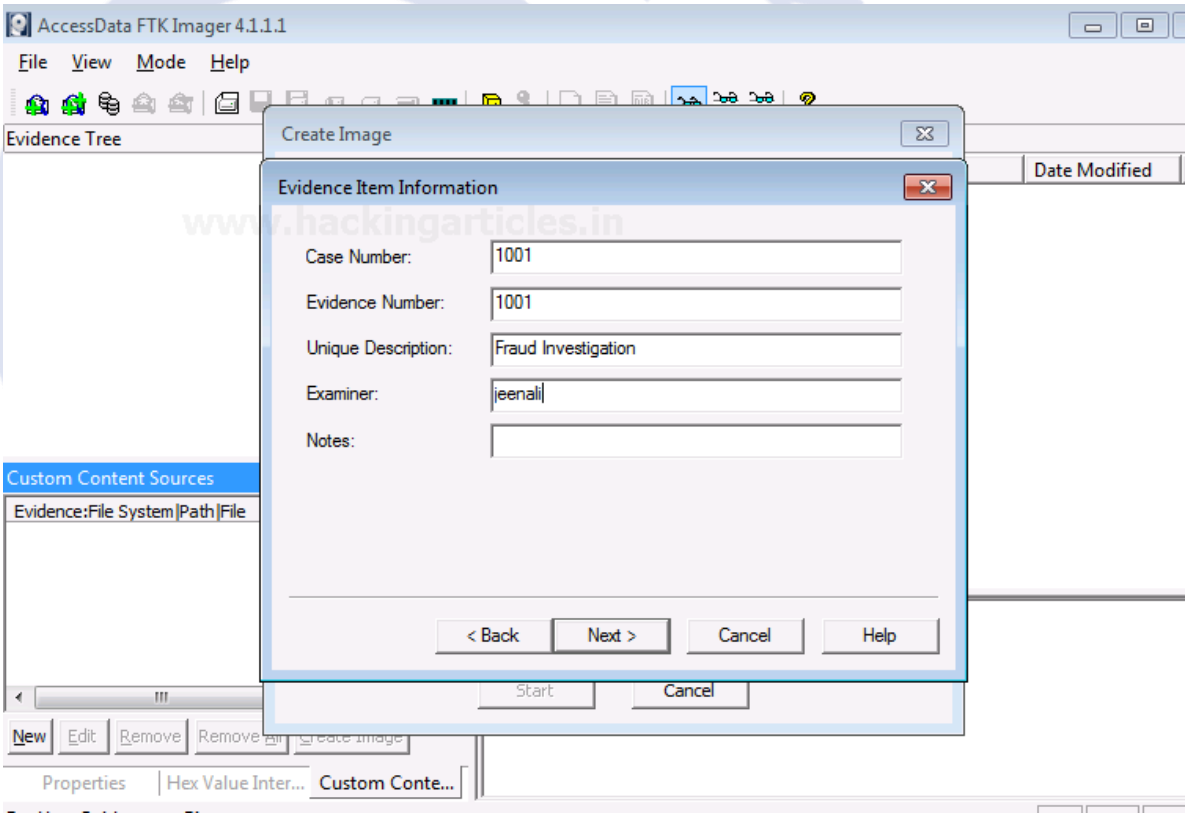
3.3 Selecting the Image Format

Select the format of the image that you want to create. The different formats for creating the image are:

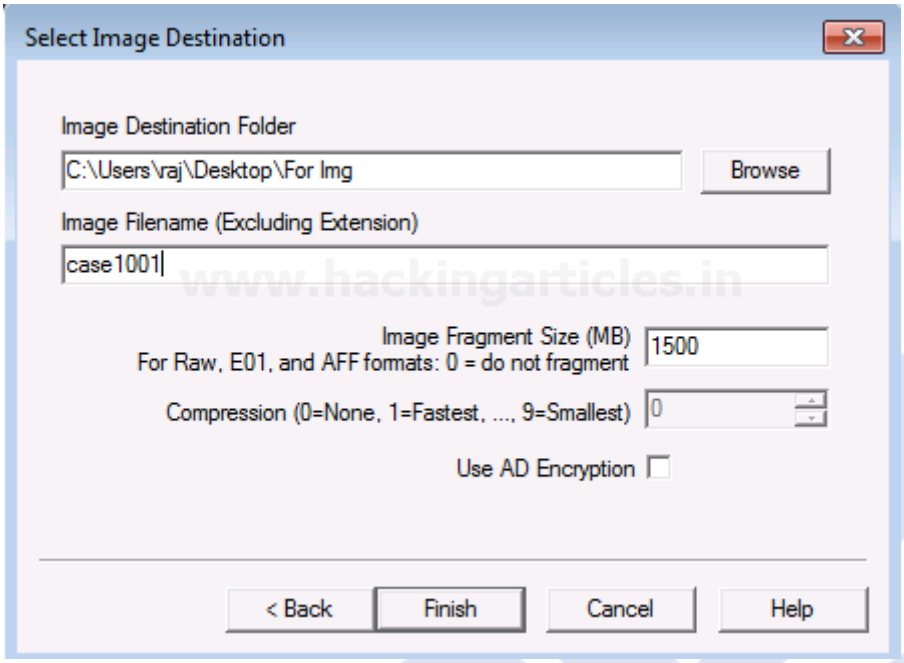
- **Raw (dd)**: A bit-by-bit copy of the original evidence is created without any additions or deletions. It does not contain any metadata.
- **SMART**: An image format that was used for Linux but has fallen out of popularity.
- **E01**: EnCase Evidence File commonly used for imaging stands for E01 and is similar to other forensic formats.
- **AFF**: Advanced Forensic Format that is an open-source format type.



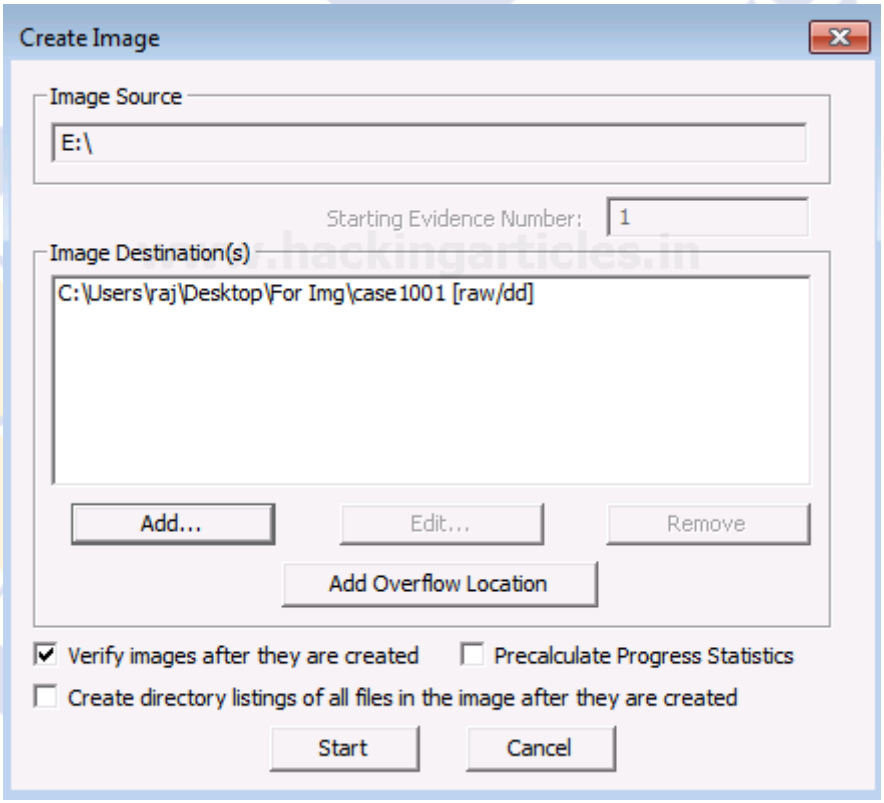
Add the details of the image to proceed.



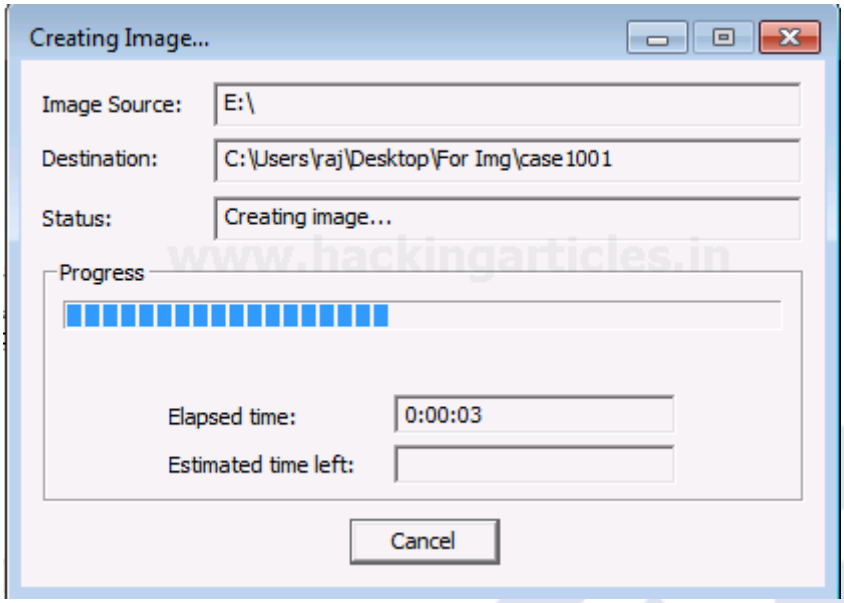
Finally, add the destination of the image file, name the image file, and then click on Finish.



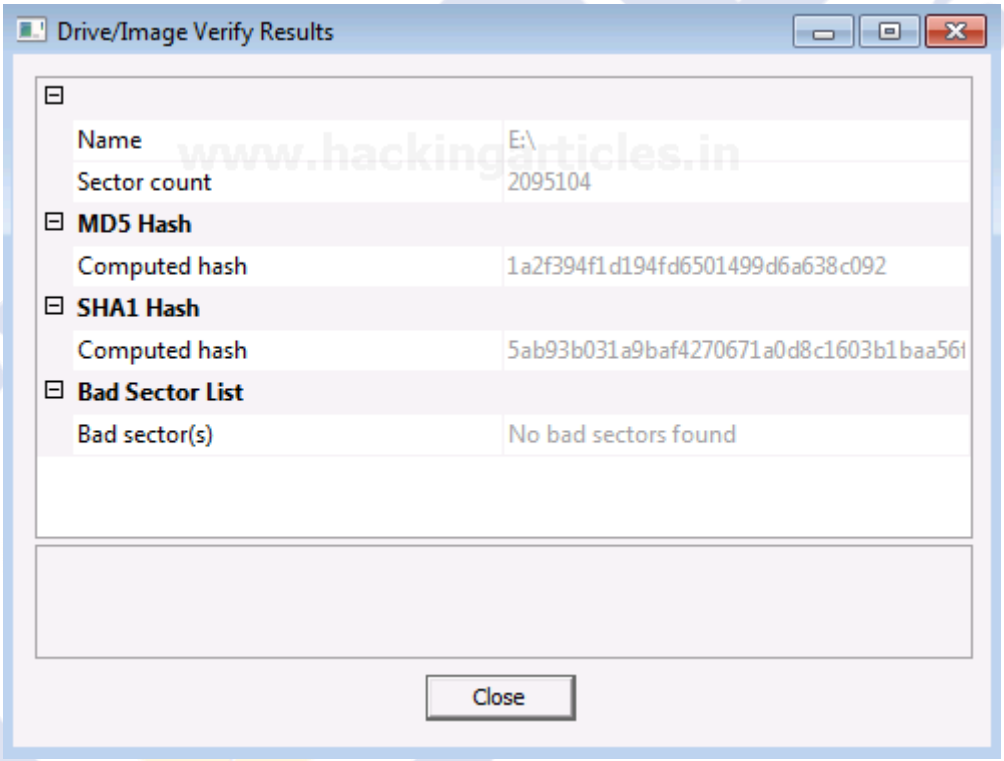
Once you have added the destination path, you can now start with the imaging and also click on the verify option to generate a hash.



Wait for a few minutes for the image to be created.



After the image is created, a hash result is generated which verifies the MD5 Hash, SHA1 Hash, and the presence of any bad sectors.



In the same directory as the forensic images, there will be a text file which contains the details about the forensic image. It contains information about the media, file hashes, and details that you entered while creating the image.

Congratulations! You have created a forensic image of a physical drive.

4. Capturing Memory

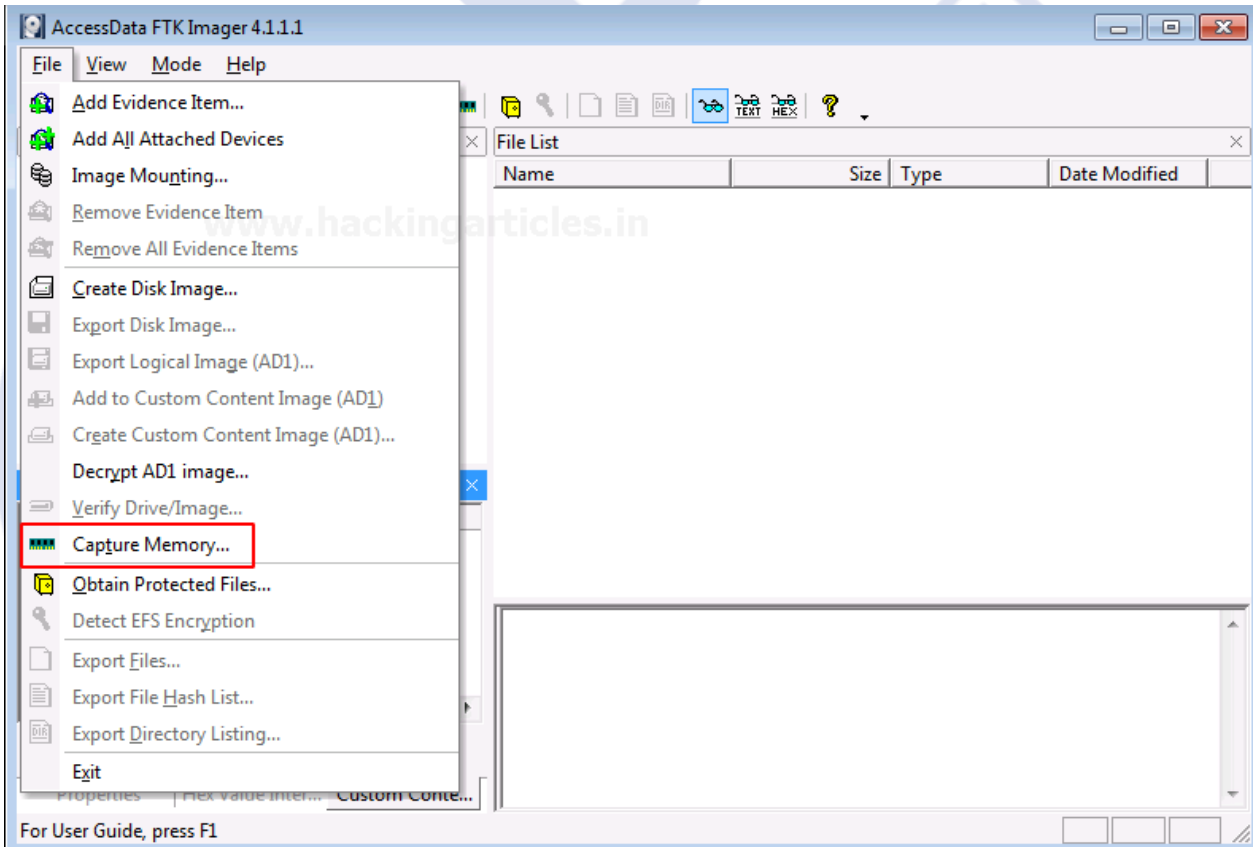
Memory capture is the method of capturing and dumping the contents of volatile content into a non-volatile storage device to preserve it for further investigation. A RAM analysis can only be successfully conducted when the acquisition has been performed accurately without corrupting the image of the volatile memory. In this phase, the investigator has to be careful about their decisions to collect volatile data as it won't exist after the system undergoes a reboot.

One feature that FTK Imager provides that tools like dd and Guymager cannot is the ability to create Memory Dumps.

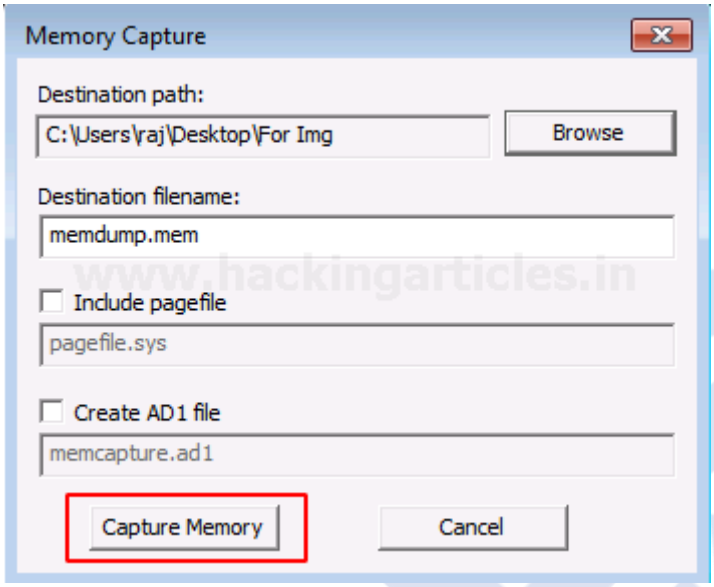
The reason why we want to capture the RAM is because a huge amount of information can be stored in the RAM without us noticing. Some information that can be stored in RAM includes:

- Temporary Data
- Encryption Keys
- Encrypted Data
- Deleted Data Fragments

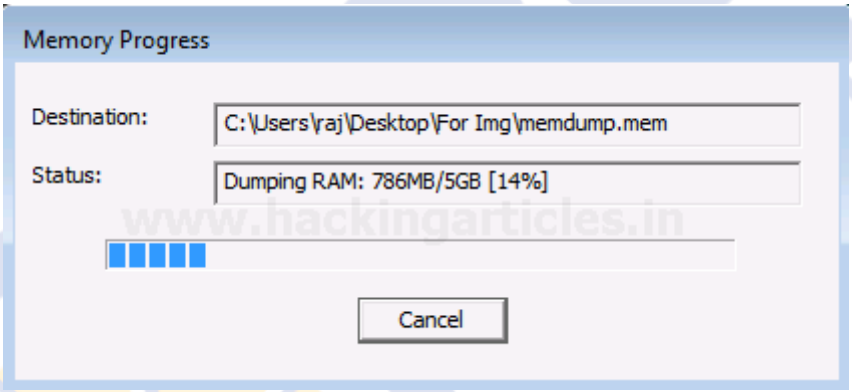
To capture the memory, click on **File > Capture Memory**.



Choose the destination path and the destination file name, and click on Capture Memory.



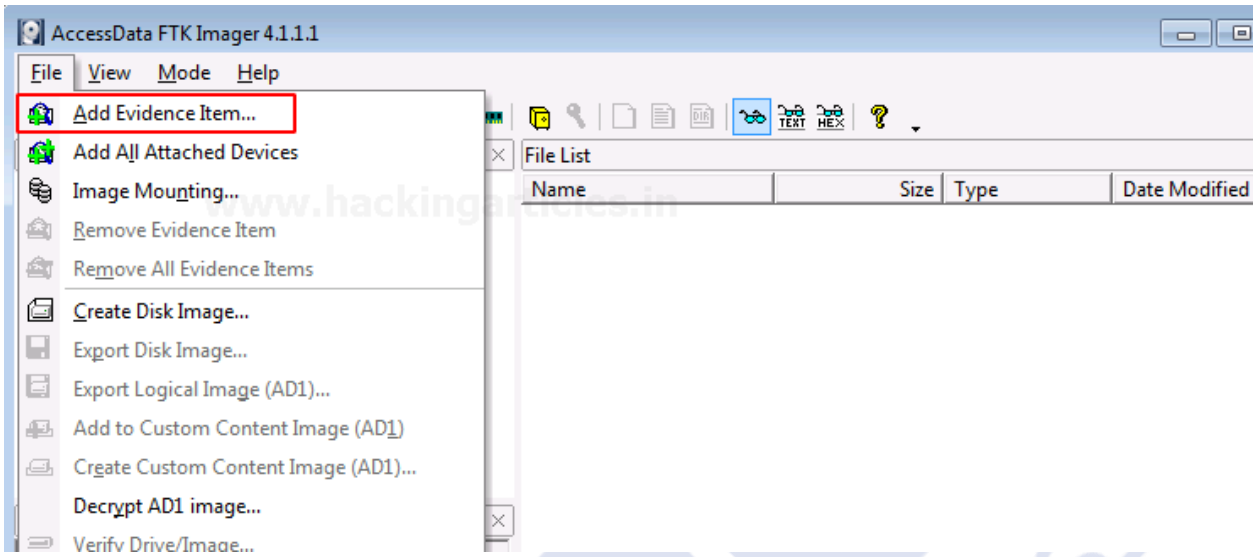
Wait for a few minutes while the RAM is being captured.



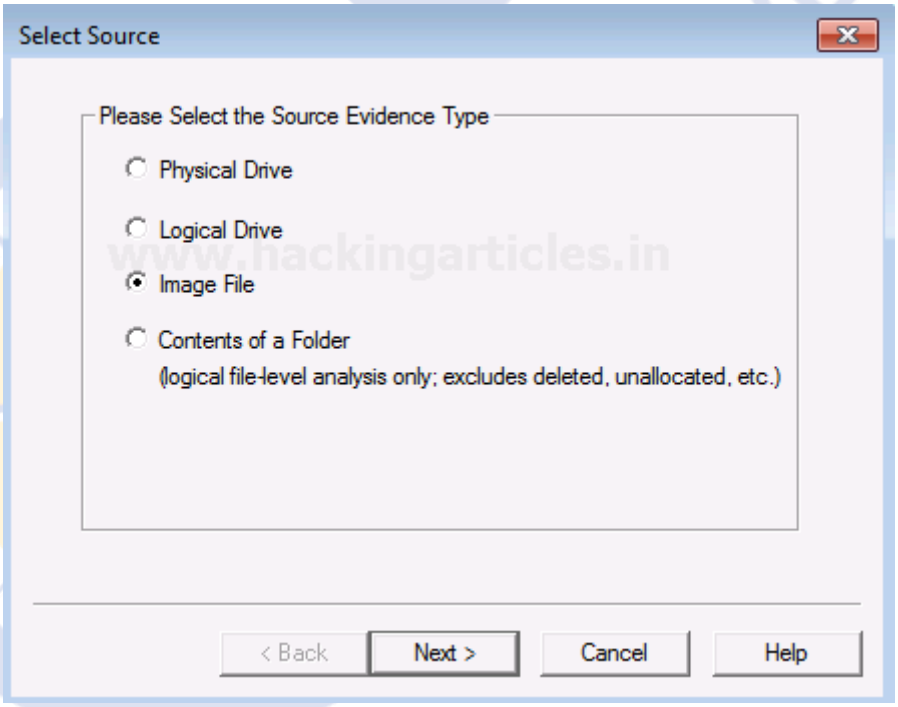
The memory dump will be saved at the destination you specified.

5. Analyzing Image Dump

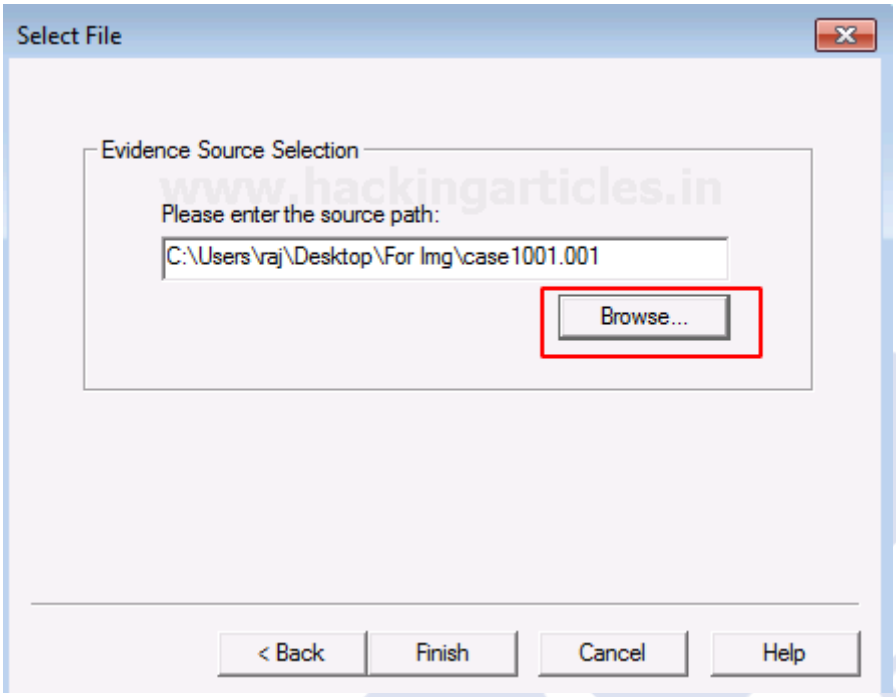
Let us analyze the Dump RAW Image once it has been acquired using FTK Imager. To start with analysis, click on **File > Add Evidence Item**.



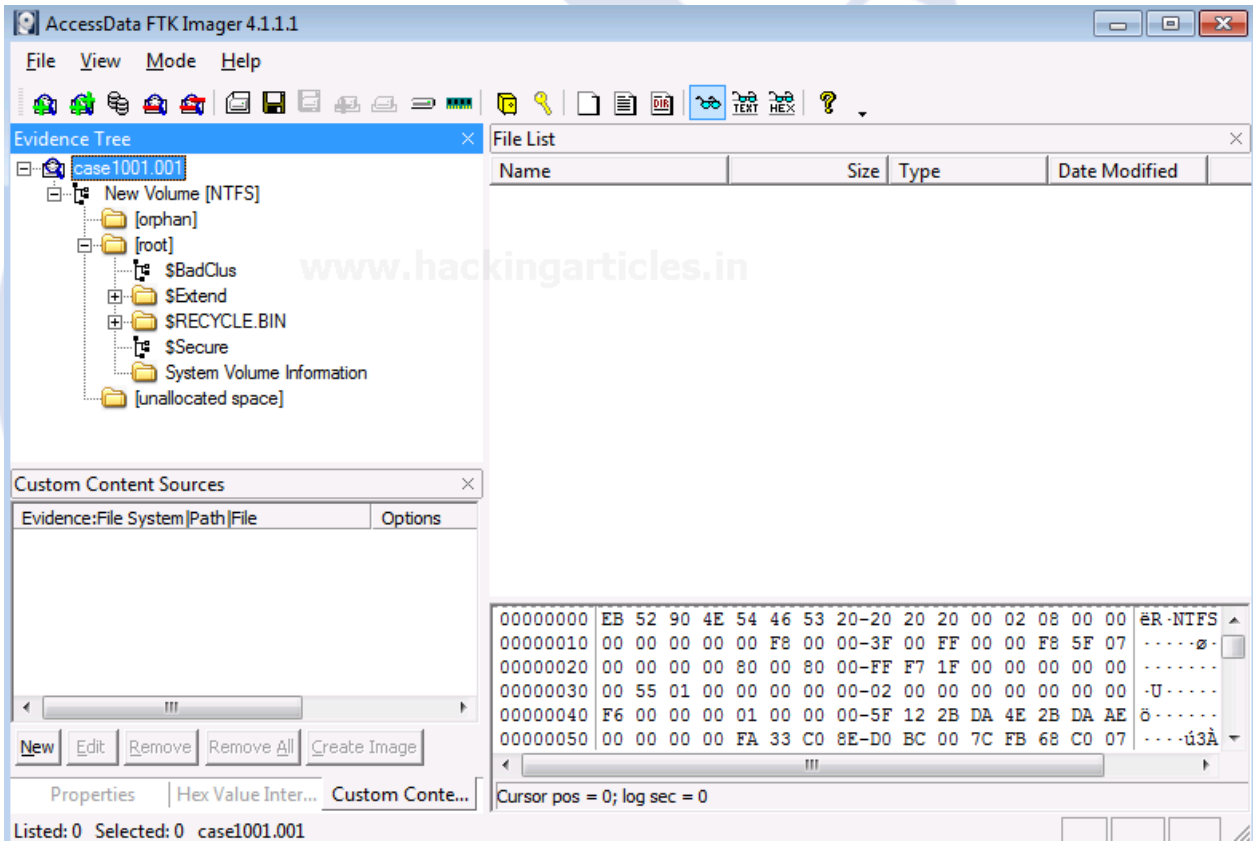
Select the source of the dump file that you have already created, so here you have to select the Image File option and click on Next.



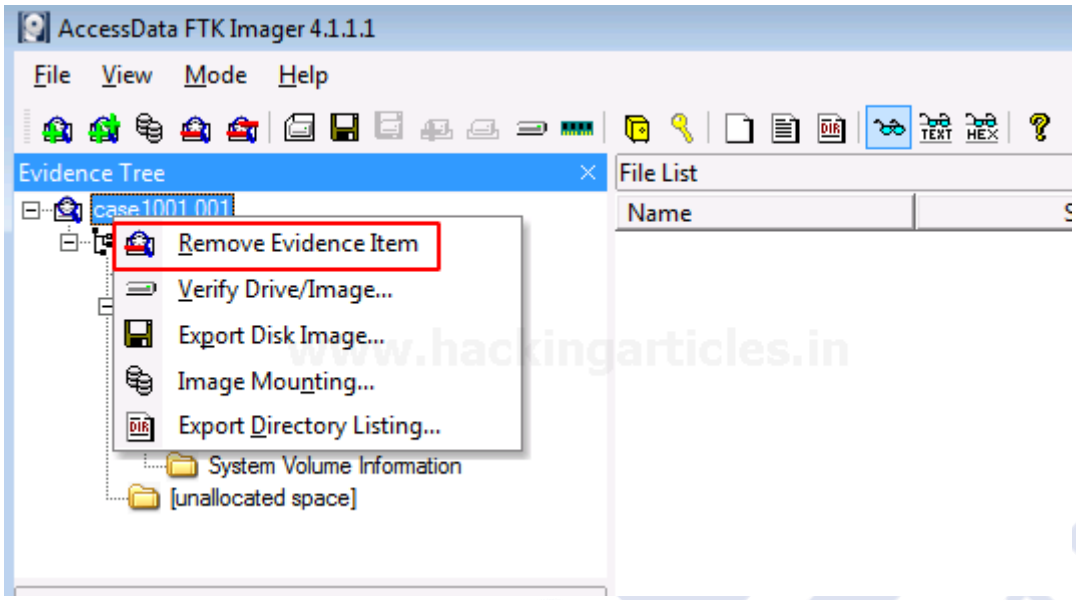
Choose the path of the image dump that you have captured by clicking on Browse.



Once you attach the image dump to the analysis part, you will see an evidence tree that contains the contents of the files of the image dump. This could include deleted as well as overwritten data.

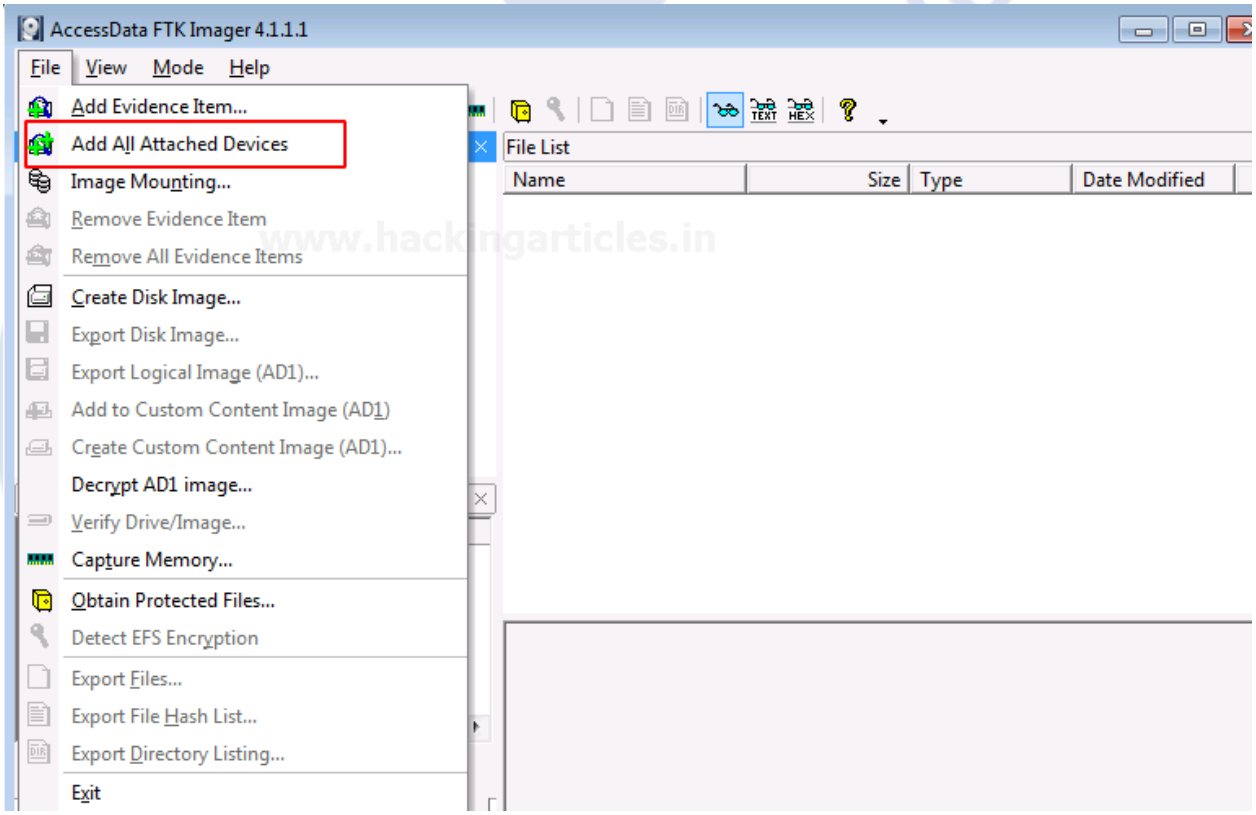


To analyze other things further, we will now remove this evidence item by right-clicking on the case and clicking on **Remove Evidence Item**.

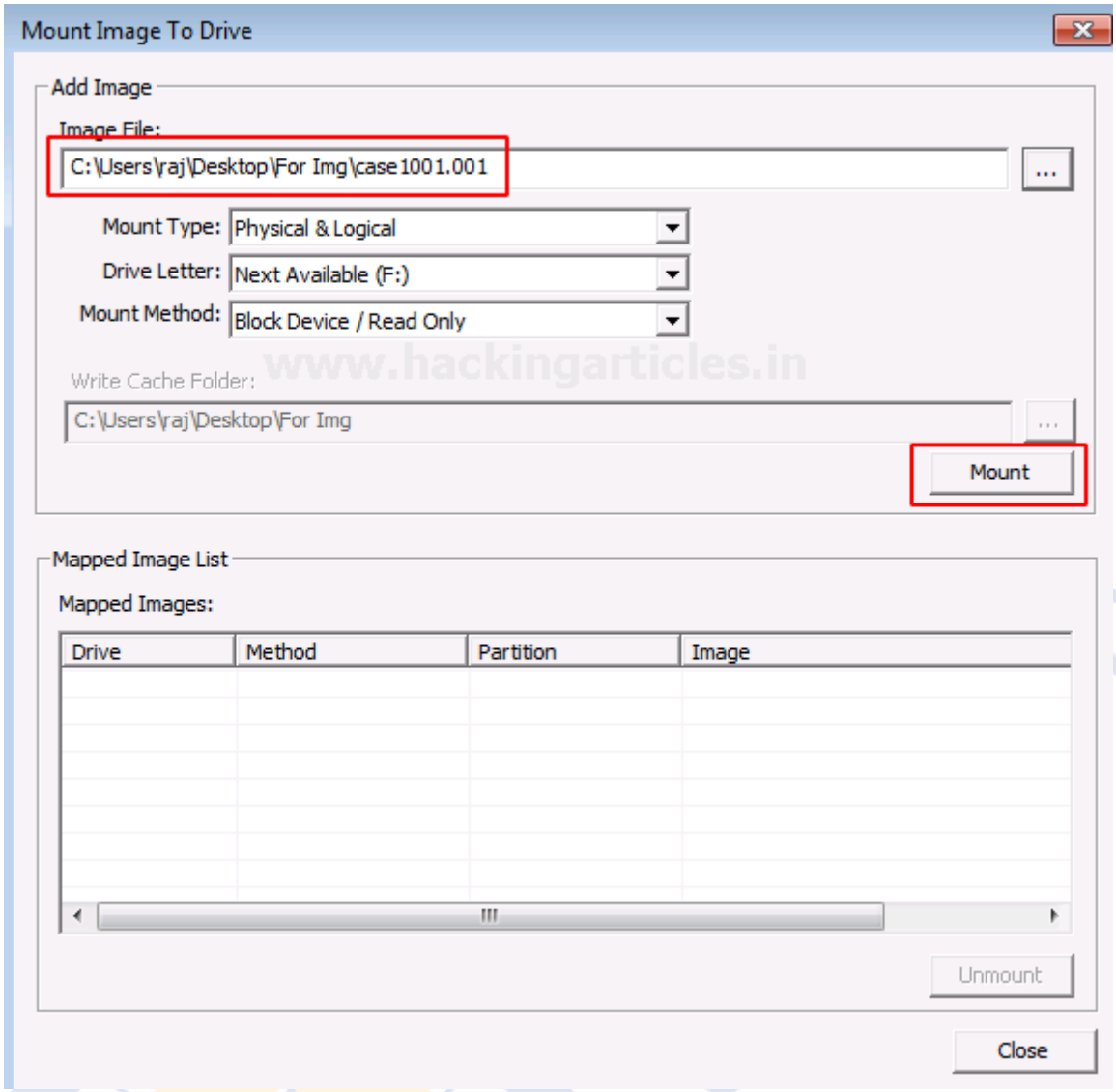


6. Mounting Image to Drive

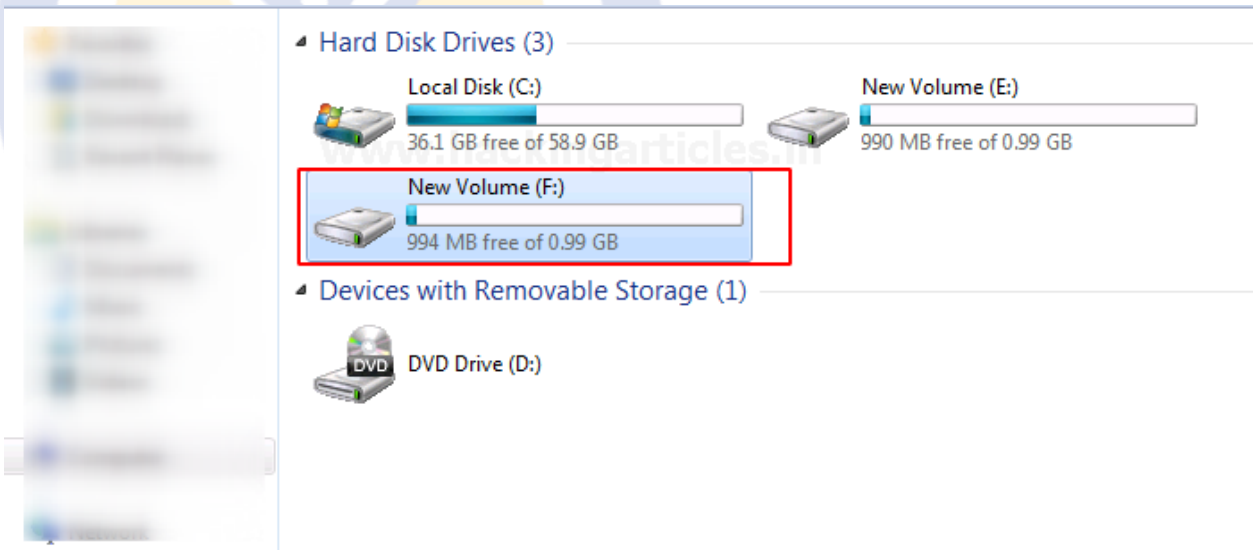
To mount the image as a drive in your system, click on **File > Image Mounting**.



Once the Mount Image to Drive window appears, you can add the path to the image file that you want to mount and click on Mount.

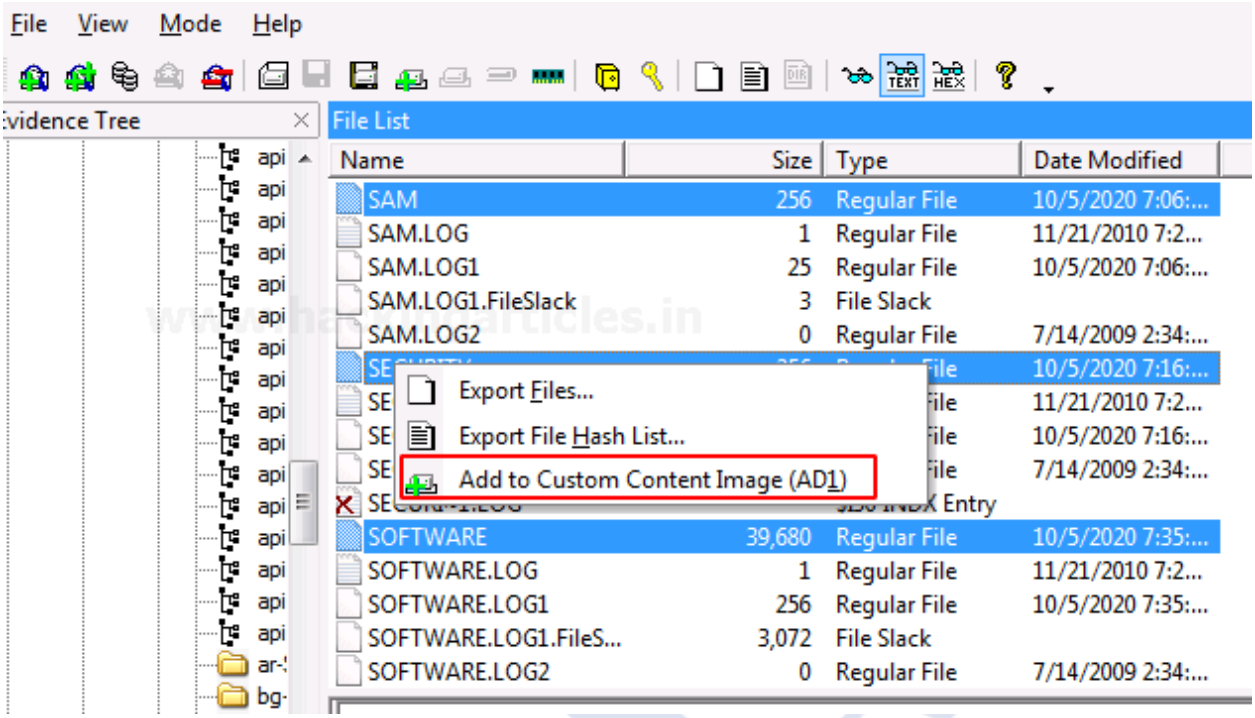


You can see that the image file has now been mounted as a drive.

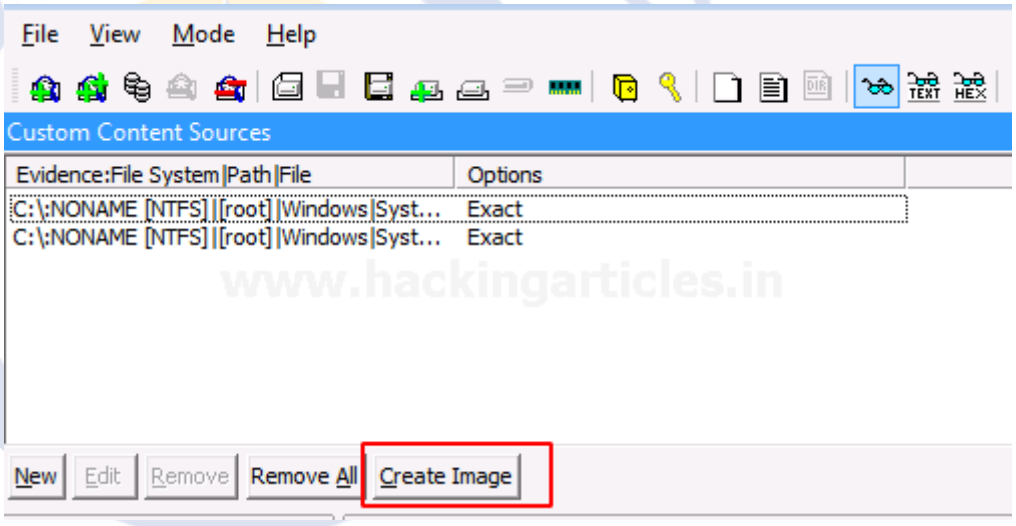


7. Custom Content Image with AD Encryption

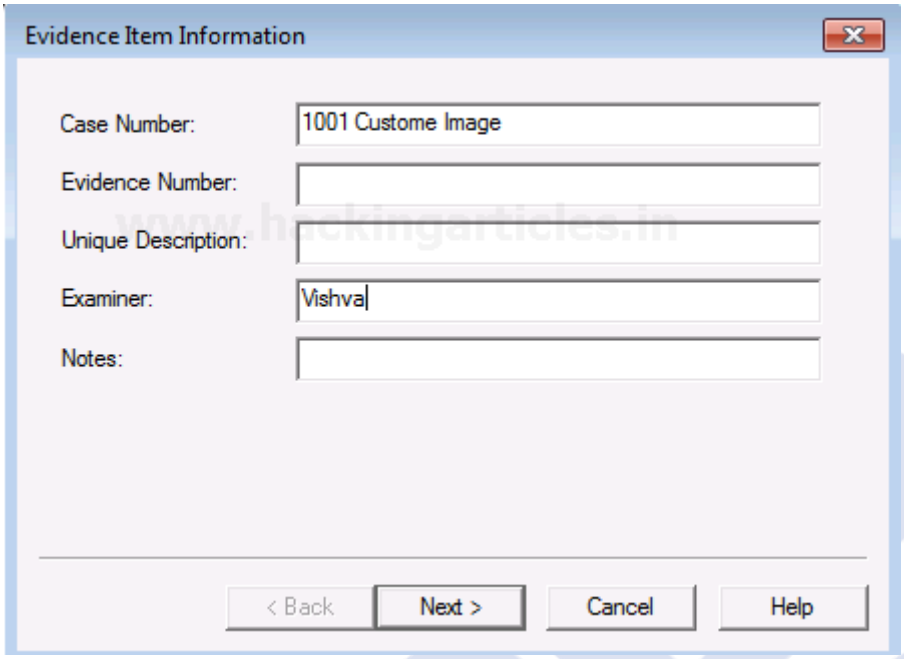
FTK Imager has a feature that allows it to encrypt files of a particular type according to the requirement of the examiner. Click on the files that you want to add to the custom content image along with AD encryption.



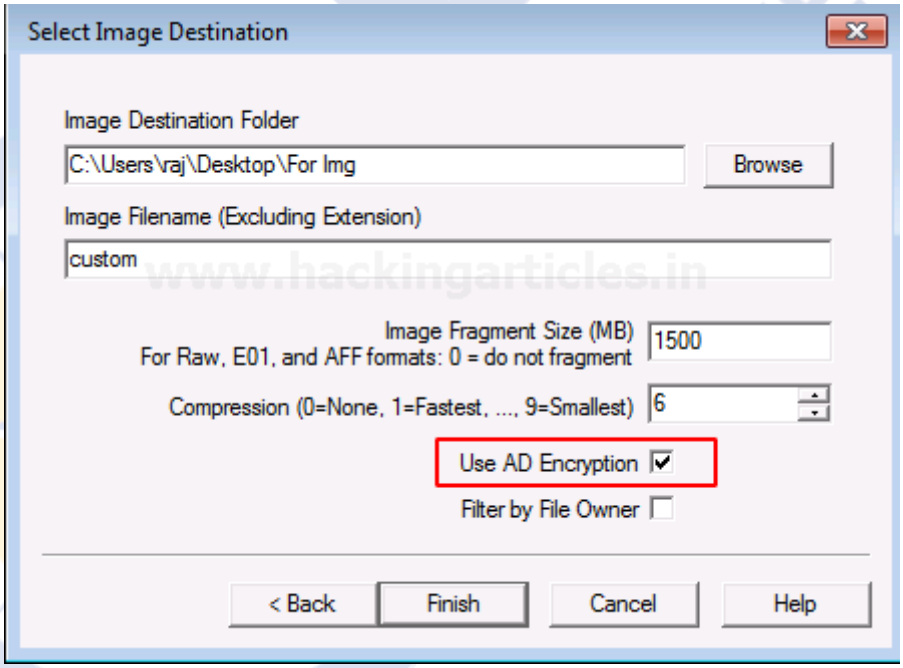
All the selected files will be displayed in a new window and then click on Create Image to proceed.



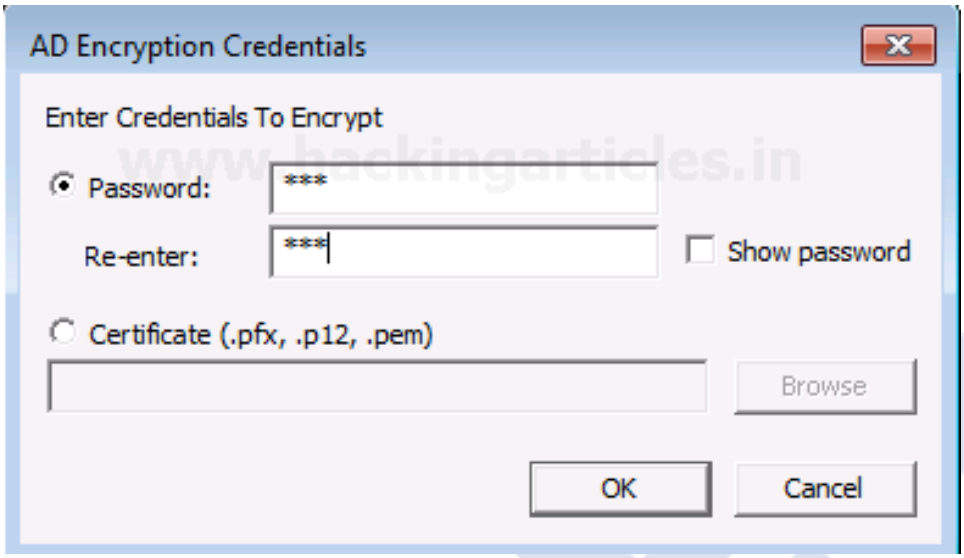
Fill in the required details to create the evidence.



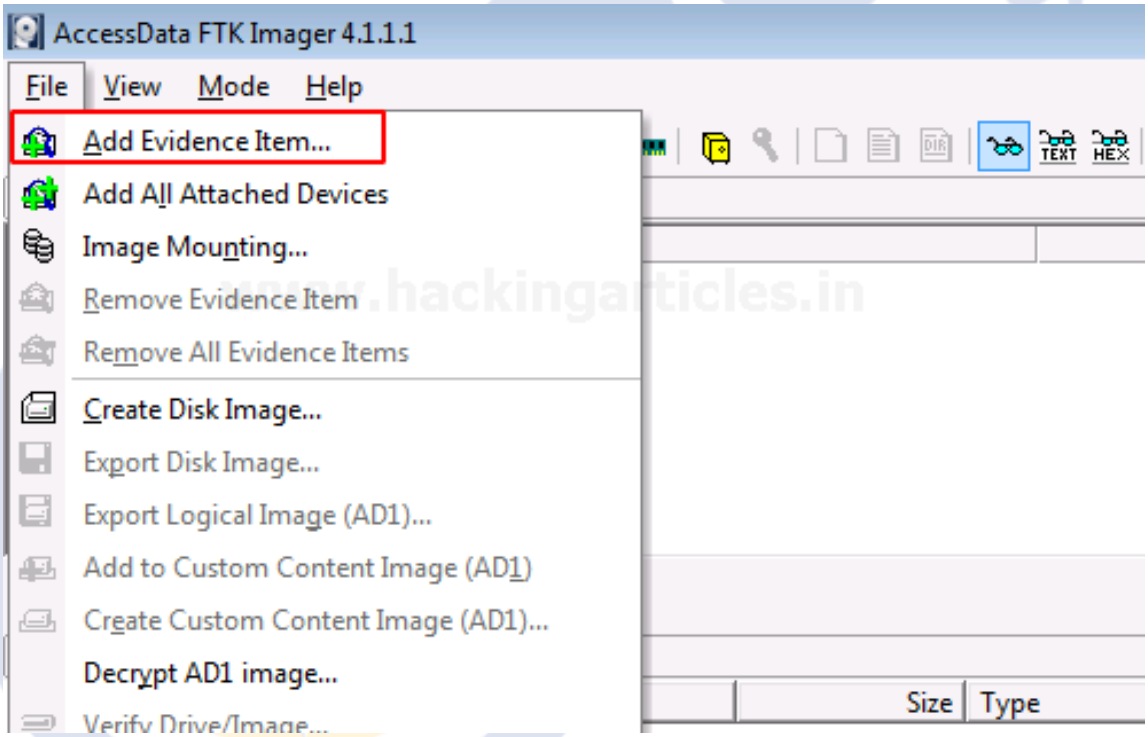
Specify the destination of the image file you want to create, name the image file, check the box with AD encryption, and then click on Finish.



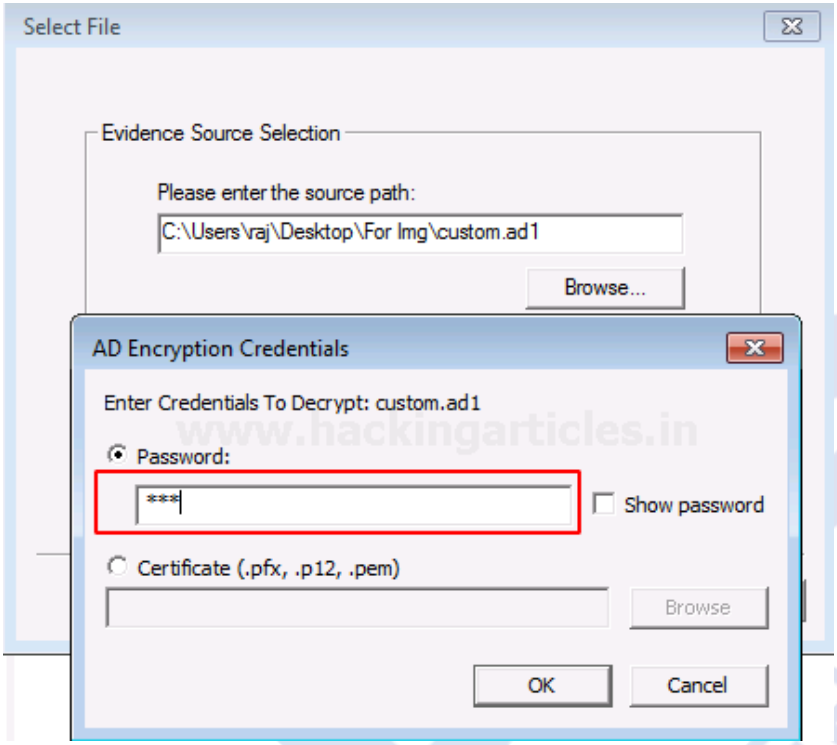
A new window will pop up to encrypt the image. Enter and re-enter the password that you want to add for your image.



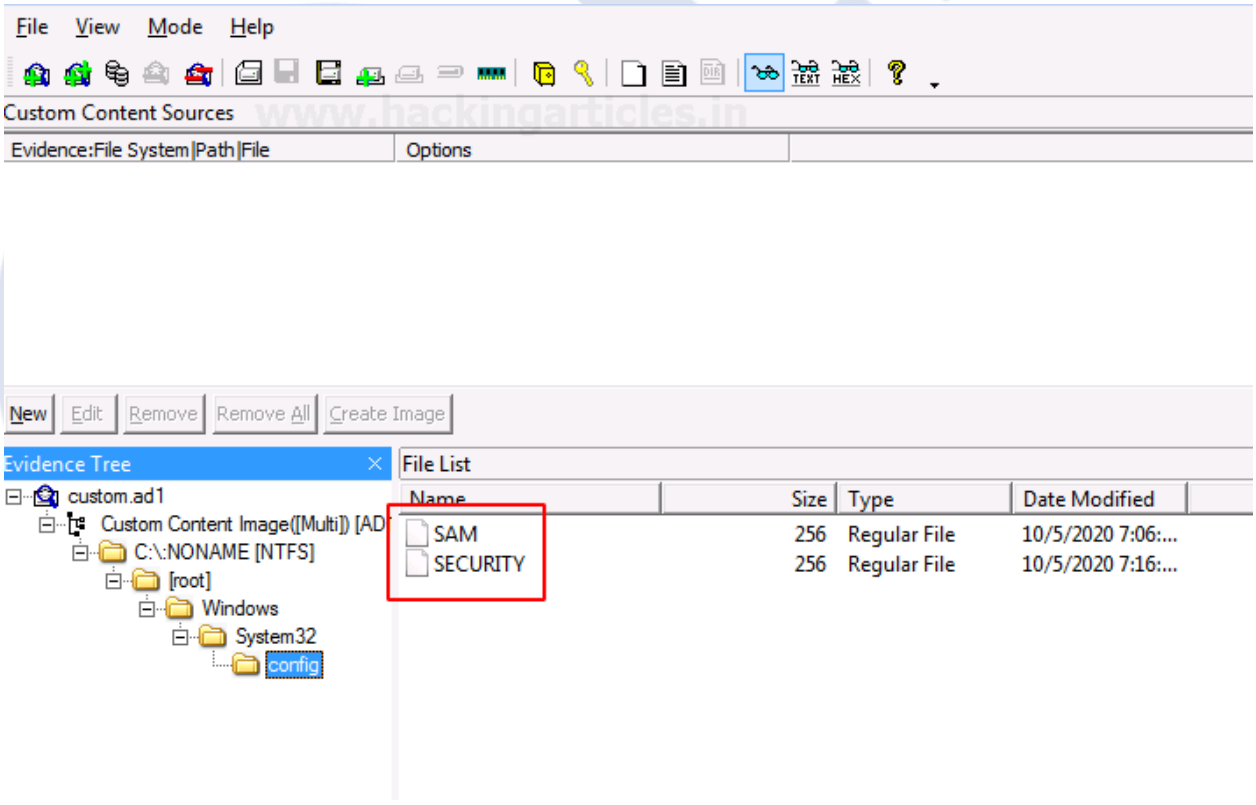
To see the encrypted files, click on **File > Add Evidence Item**.



The window to decrypt the encrypted files will appear once you add the file source. Enter the password and click OK.

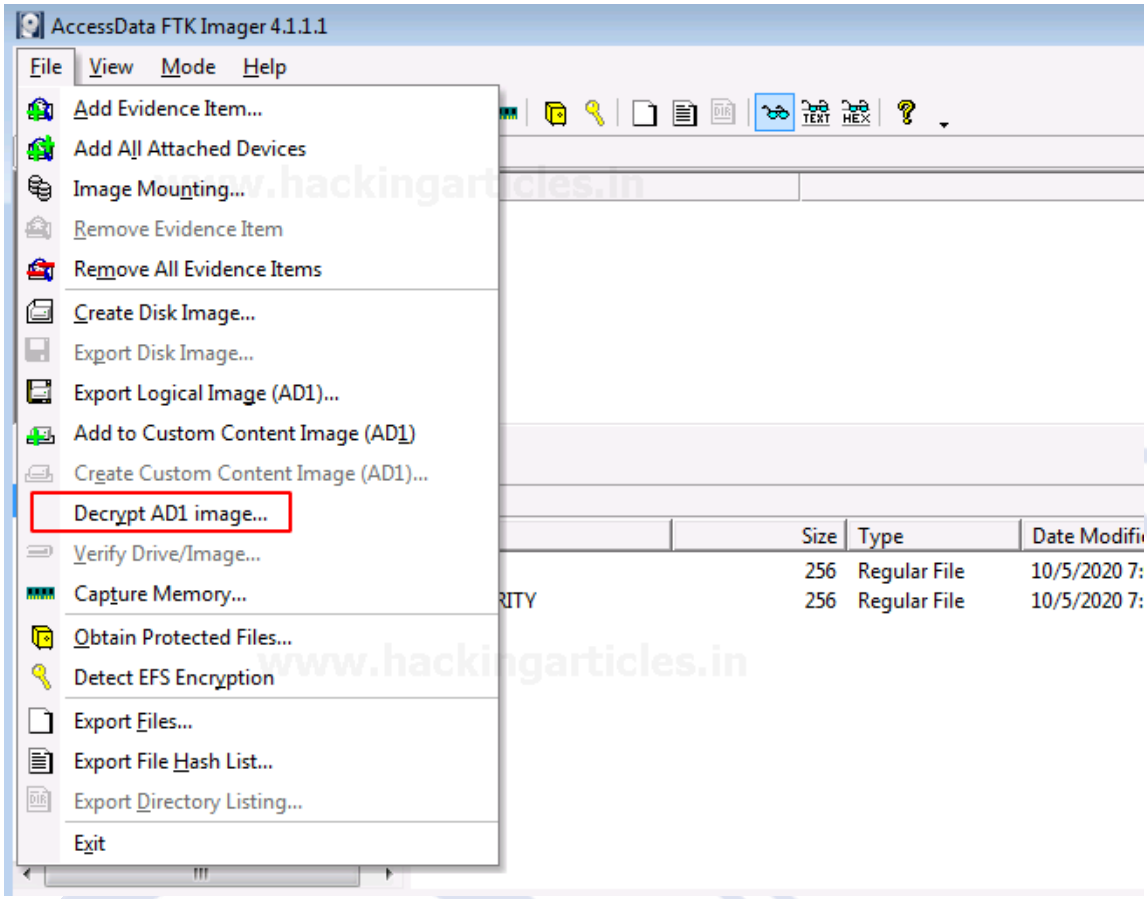


You will now see the encrypted files after entering the valid password.

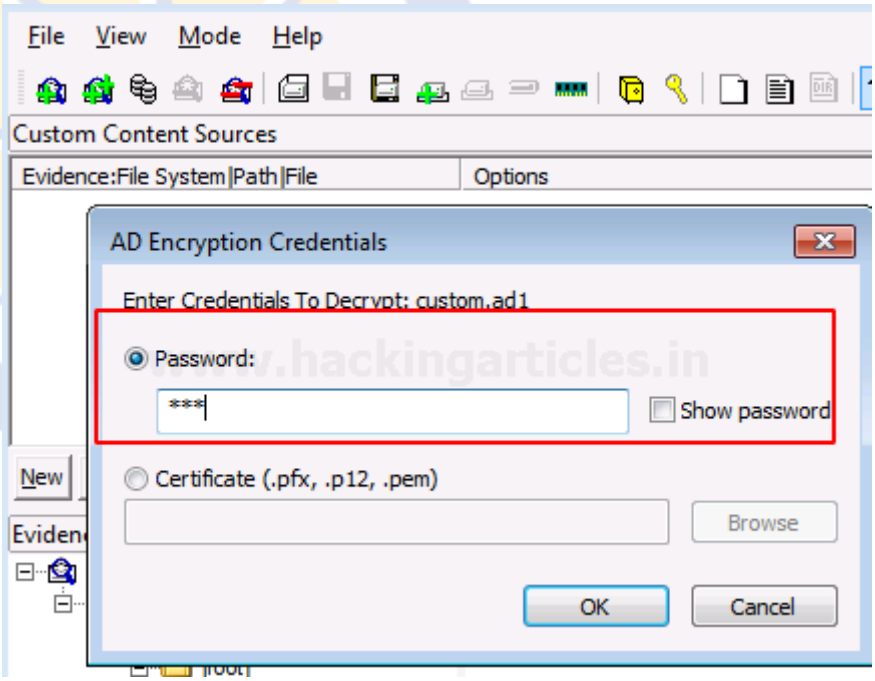


8. Decrypt AD1 Image

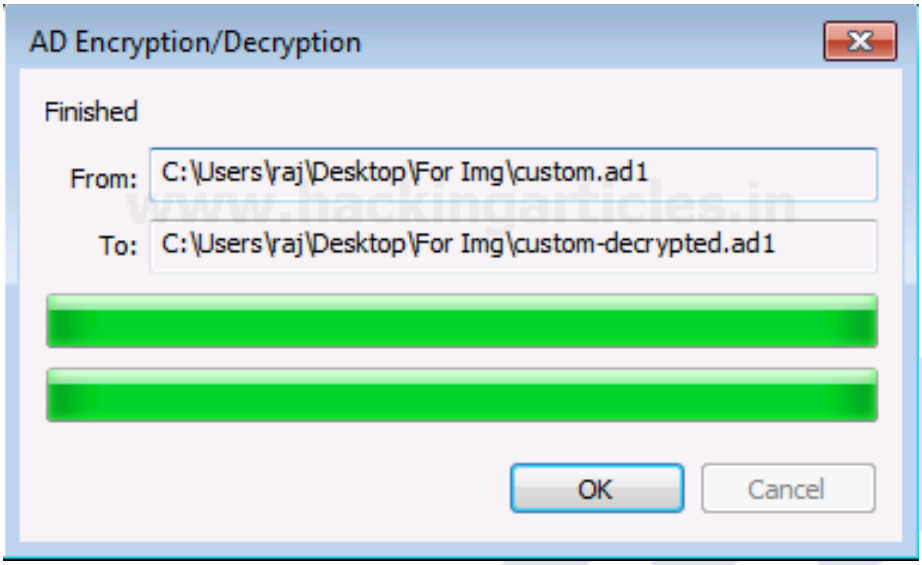
To decrypt the custom content image, click on **File > Decrypt AD1 Image**.



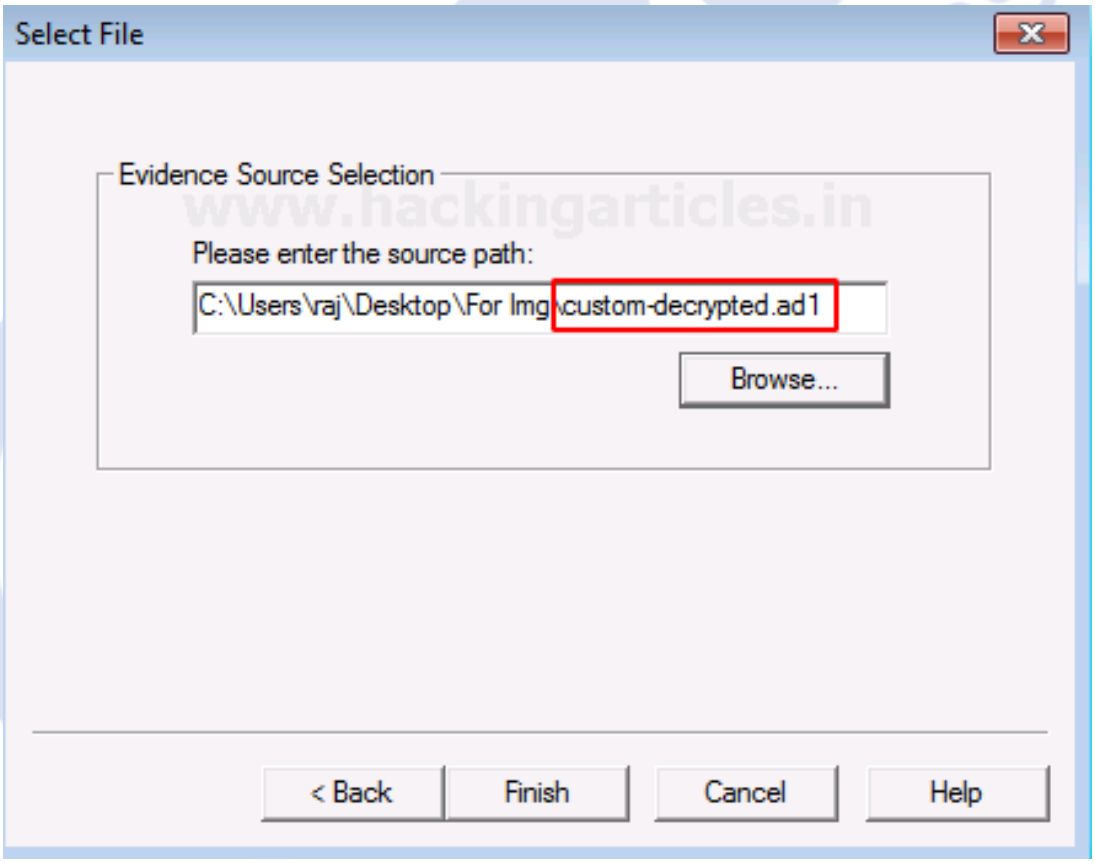
You need to enter the password for the image file that was encrypted and click on OK.



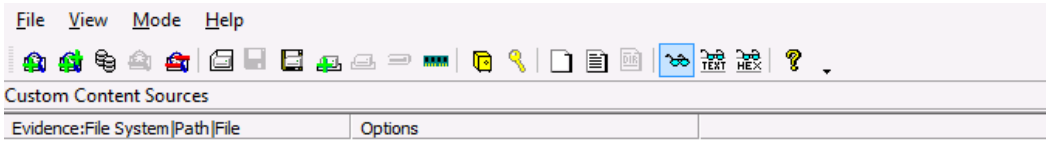
Wait for a few minutes until the decrypted image is created.

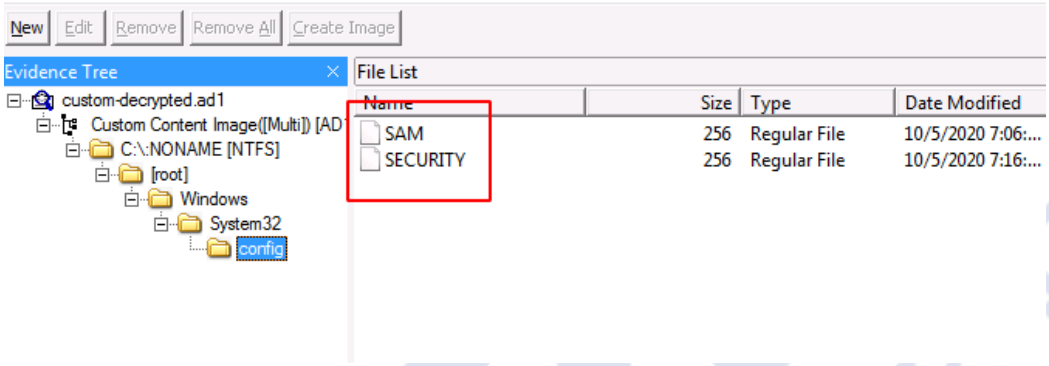


To view the decrypted custom content image, add the path of the decrypted file and click on Finish.



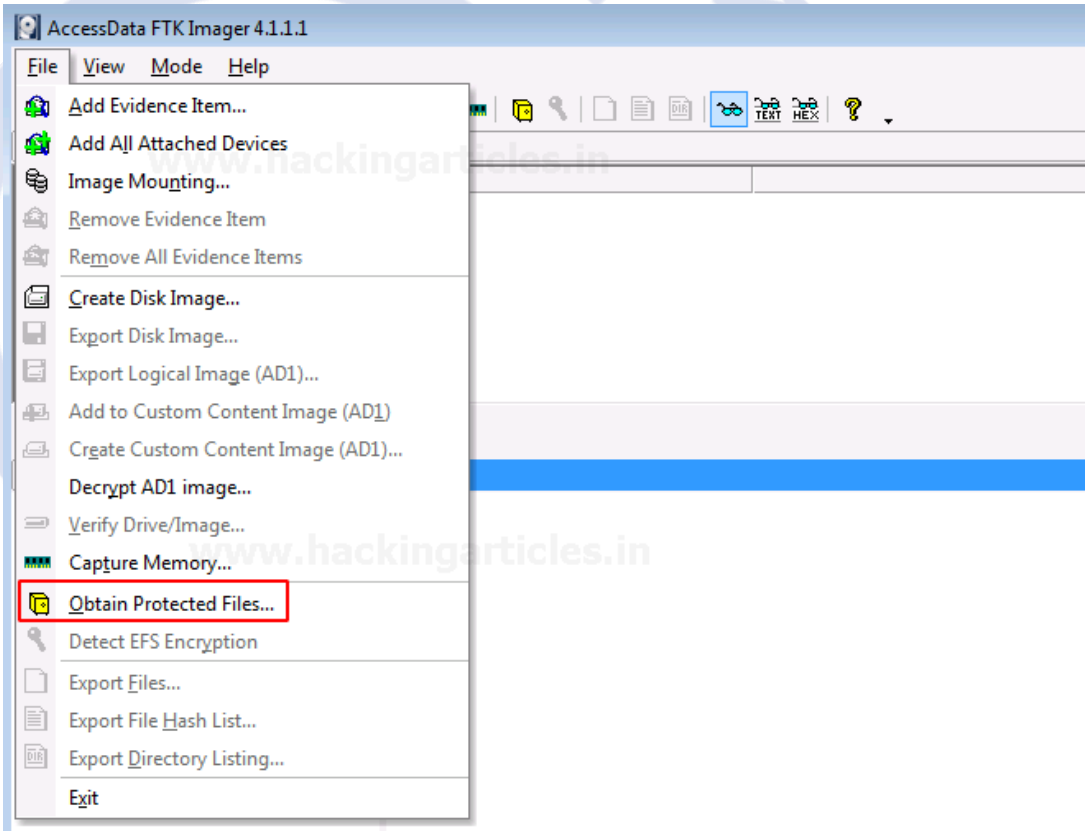
You will now be able to see the encrypted files by using the correct password to decrypt them.



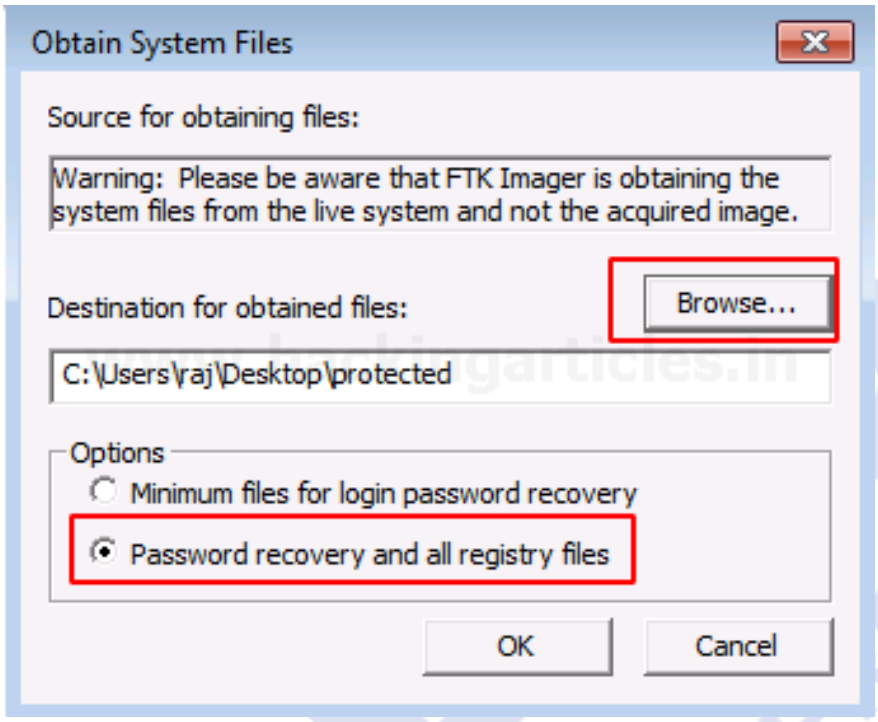


9. Obtain Protected Files

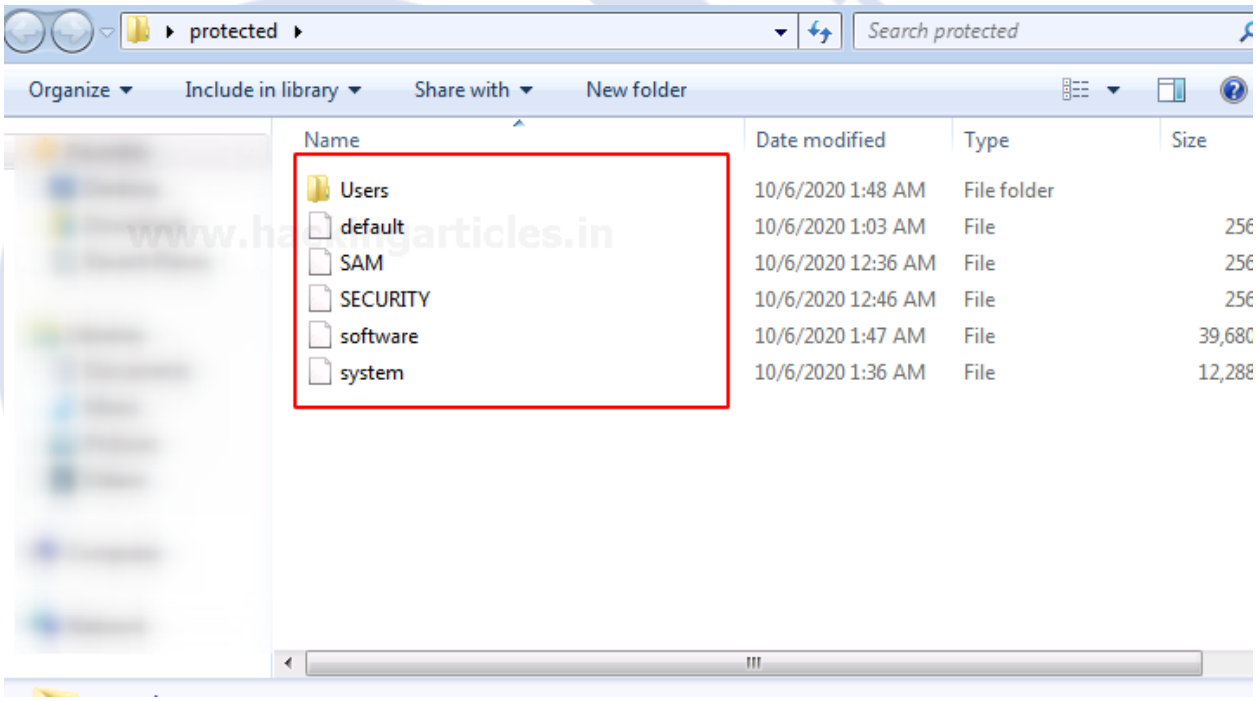
Certain files are protected on recovery. To obtain those files, click on **File > Obtain Protected Files**.



A new window will pop up. Click on Browse to add the destination of the protected file, then click on the option that says **Password Recovery and All Registry Files**, and finally click on OK.



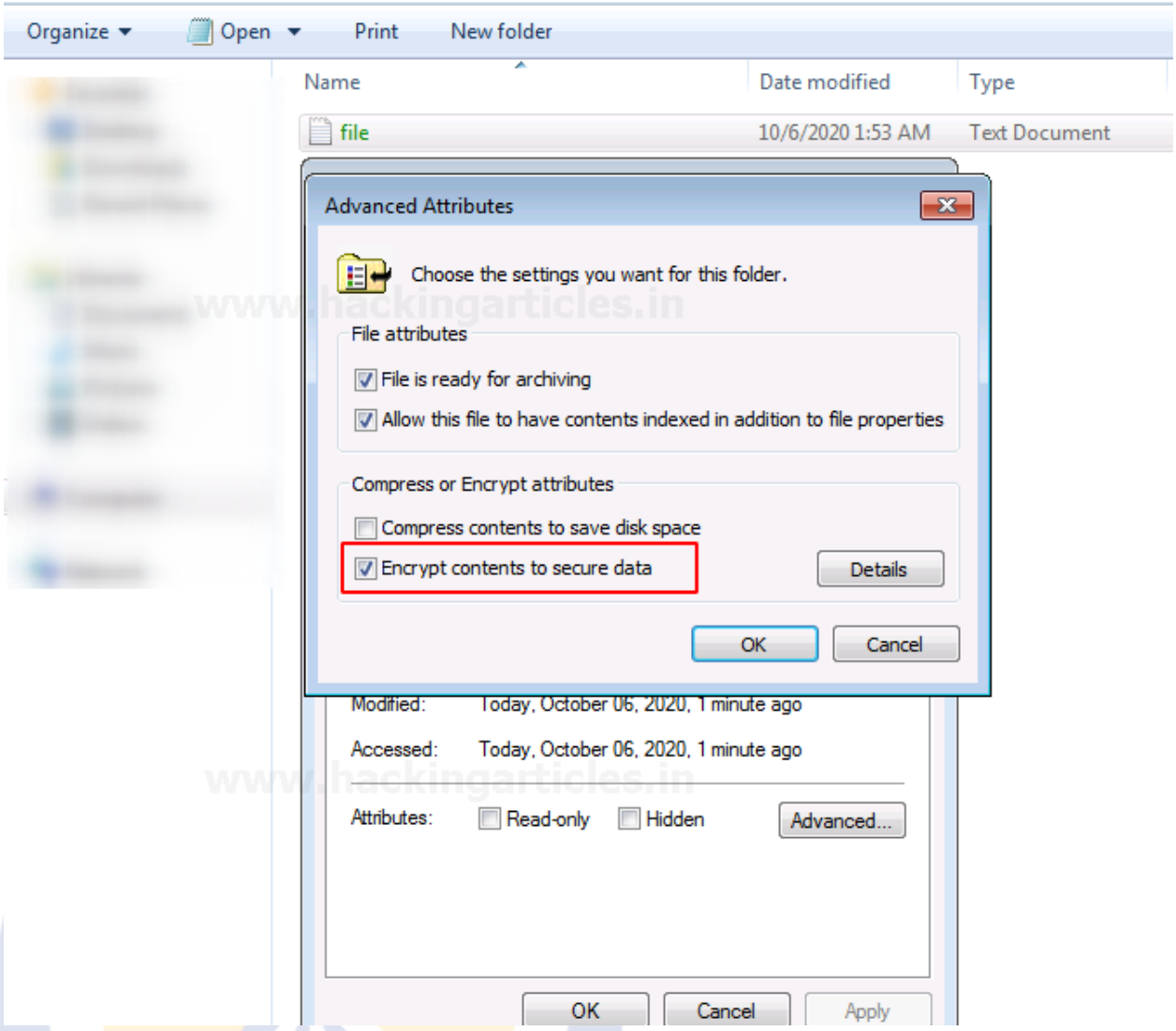
You will see all the protected files in one place.



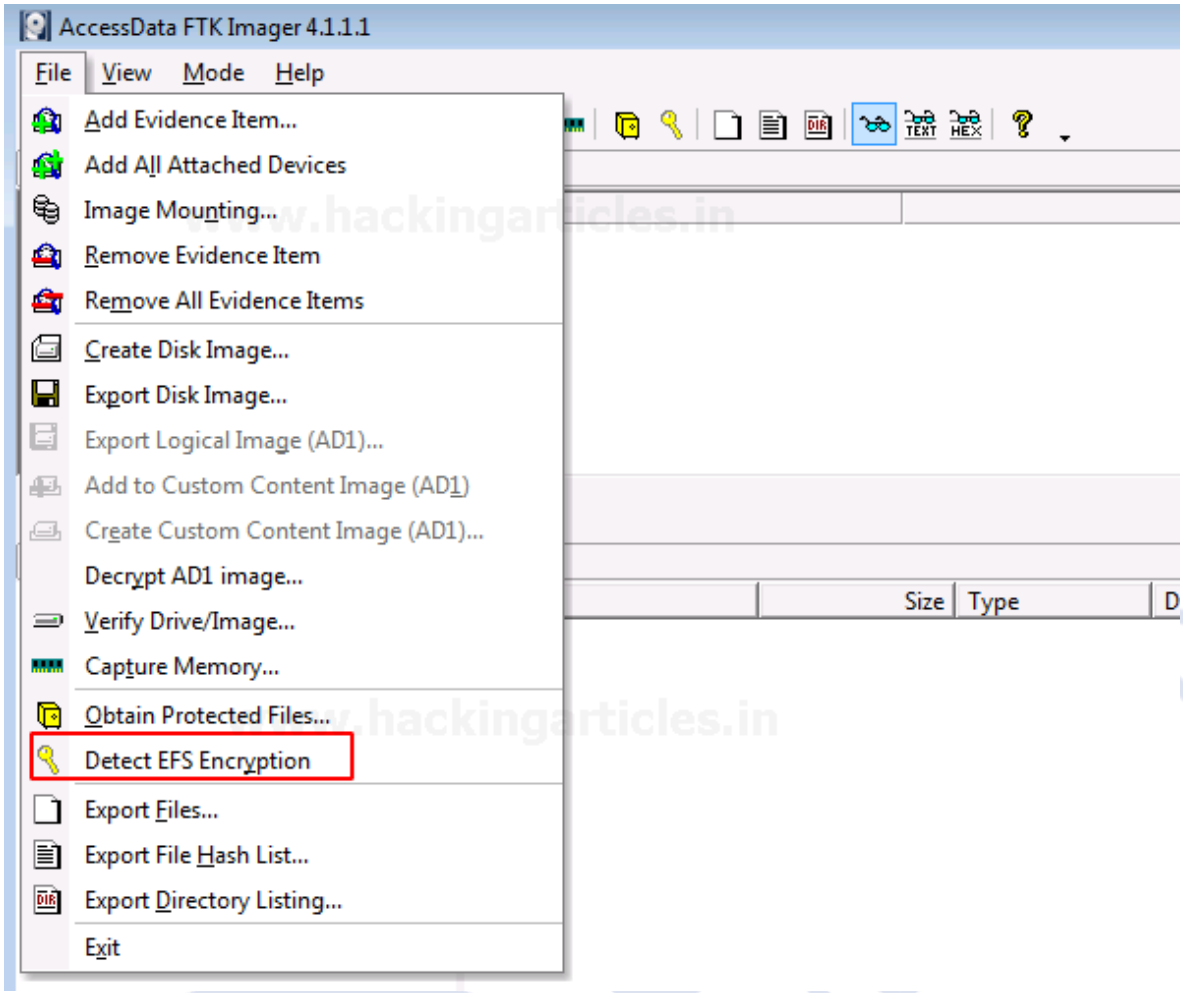
10. Detect EFS Encryption

When we encrypt a folder or a file, we can detect it using this feature of FTK Imager.

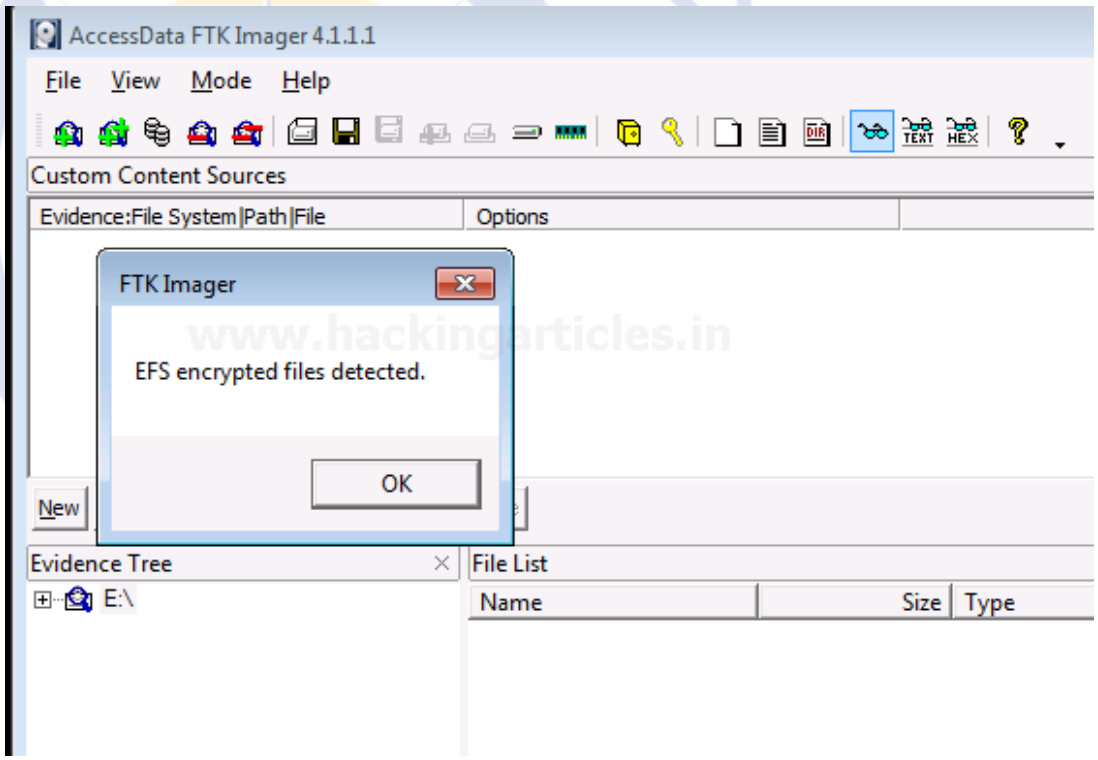
A user encrypts a file in a folder to secure its content.



To detect the EFS encryption, click on **File > Detect EFS Encryption**.

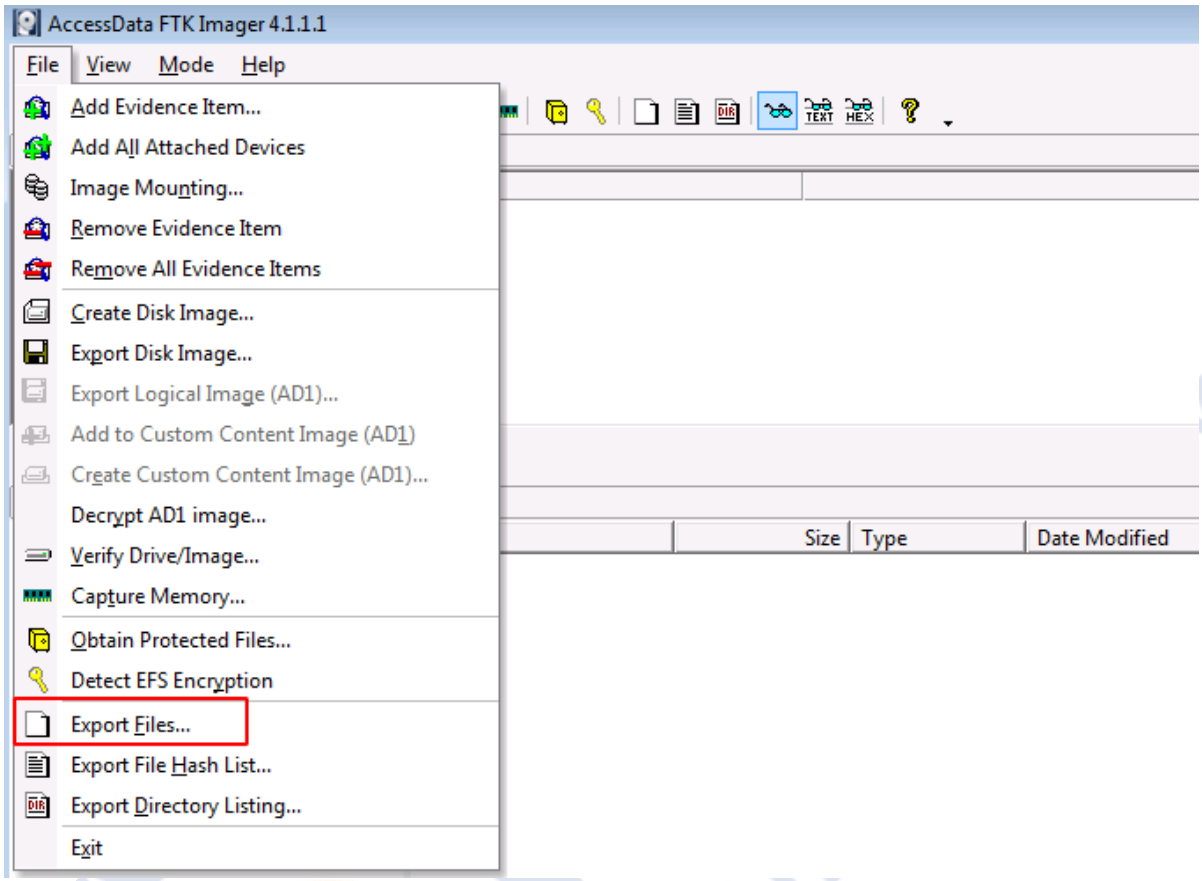


You can see that the encryption is detected.

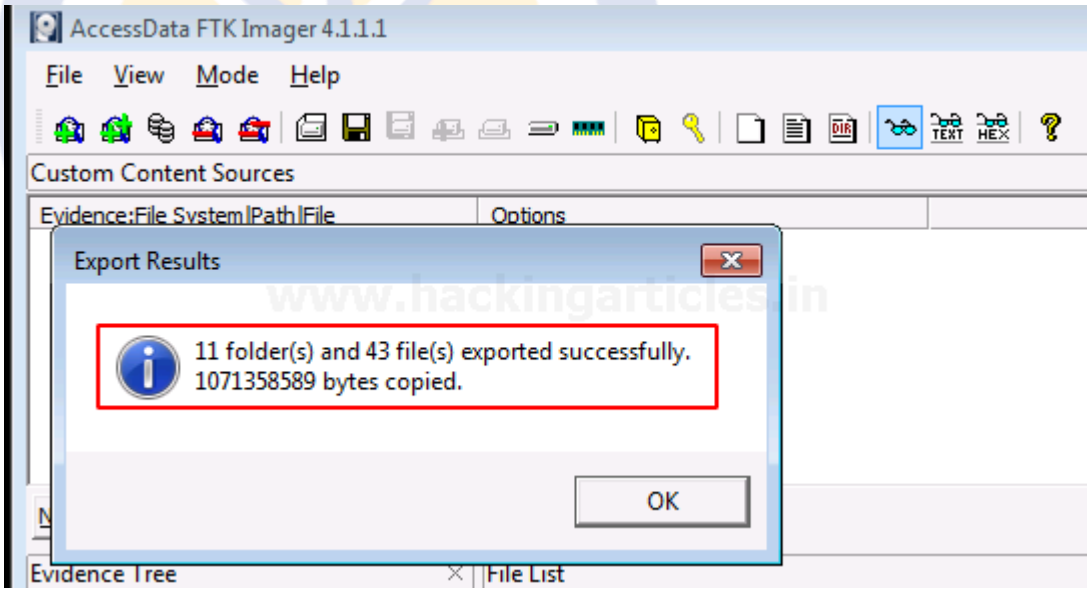


11. Export Files

To export the files and folders from the imaged file to your folder, click on **File > Export Files**.



You can now see the results of the export showing the number of files and folders that the system has copied.



12. Disadvantages

While FTK Imager is a powerful tool, it has some limitations:

Windows-Only Software

- It is unfortunate that Linux is not supported, as Linux is often the chosen OS for creating these kinds of investigative environments.

Slow Performance with Large Datasets

- When a physical drive that is very large needs to be imaged, it will take a long time as it is not optimized for handling high-volume data.

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13. Conclusion

FTK Imager is an essential tool in the digital forensics toolkit. It provides comprehensive features for creating forensic images, capturing memory, analyzing evidence, and maintaining the integrity of digital evidence through hash verification. Understanding how to properly use FTK Imager is crucial for anyone involved in digital forensics and incident response.

The tool's ability to create bit-by-bit copies, capture volatile memory, mount images for analysis, and maintain chain of custody through hash verification makes it invaluable for forensic investigators. Despite its limitations as Windows-only software and performance issues with very large datasets, FTK Imager remains one of the most widely used tools in digital forensics.

FTK Imager – Lab Tasks & Questions

SECTION A: CONCEPTUAL TASKS

Task 1: Logical vs Physical Disk Imaging
What distinguishes a logical disk image from a physical disk image in FTK Imager? Why would you choose to acquire a logical disk image instead of a physical one?

Task 2: Image Integrity Verification
Did you verify the integrity of the image? If so, how did you perform the verification?

Task 3: DD Command for Disk Imaging
What is the purpose of using the dd command for disk imaging, and how does it compare to other imaging tools like FTK Imager?

Task 4: Guymager Disk Image Formats
What types of disk image formats does Guymager support, and how did you choose the most appropriate format for your case?

Task 5: Documentation and Screenshots
Provide screenshots of the disk image you obtained, along with supporting documentation that includes details about the tool used and the type of acquisition (physical or logical) performed.

Task 6: Insights and Importance
What insights did you gain from completing the disk acquisition tasks in this lab? Why is the process of acquiring and creating disk images crucial in digital forensics, and how does it ensure the integrity and validity of digital evidence?

SECTION B: TECHNICAL QUESTIONS

(As per the forensic image created by the student)

1. What was the size/capacity of the original media as identified by FTK Imager?
2. How many sectors were on the original media as identified by FTK Imager?
3. What was the serial number of the original drive as identified by FTK Imager?
4. What were the MD5 and SHA1 hash values of the original evidence file?
5. Was the forensic image an exact copy of the original media?